

BFS Traversal

Step 1: Start at A

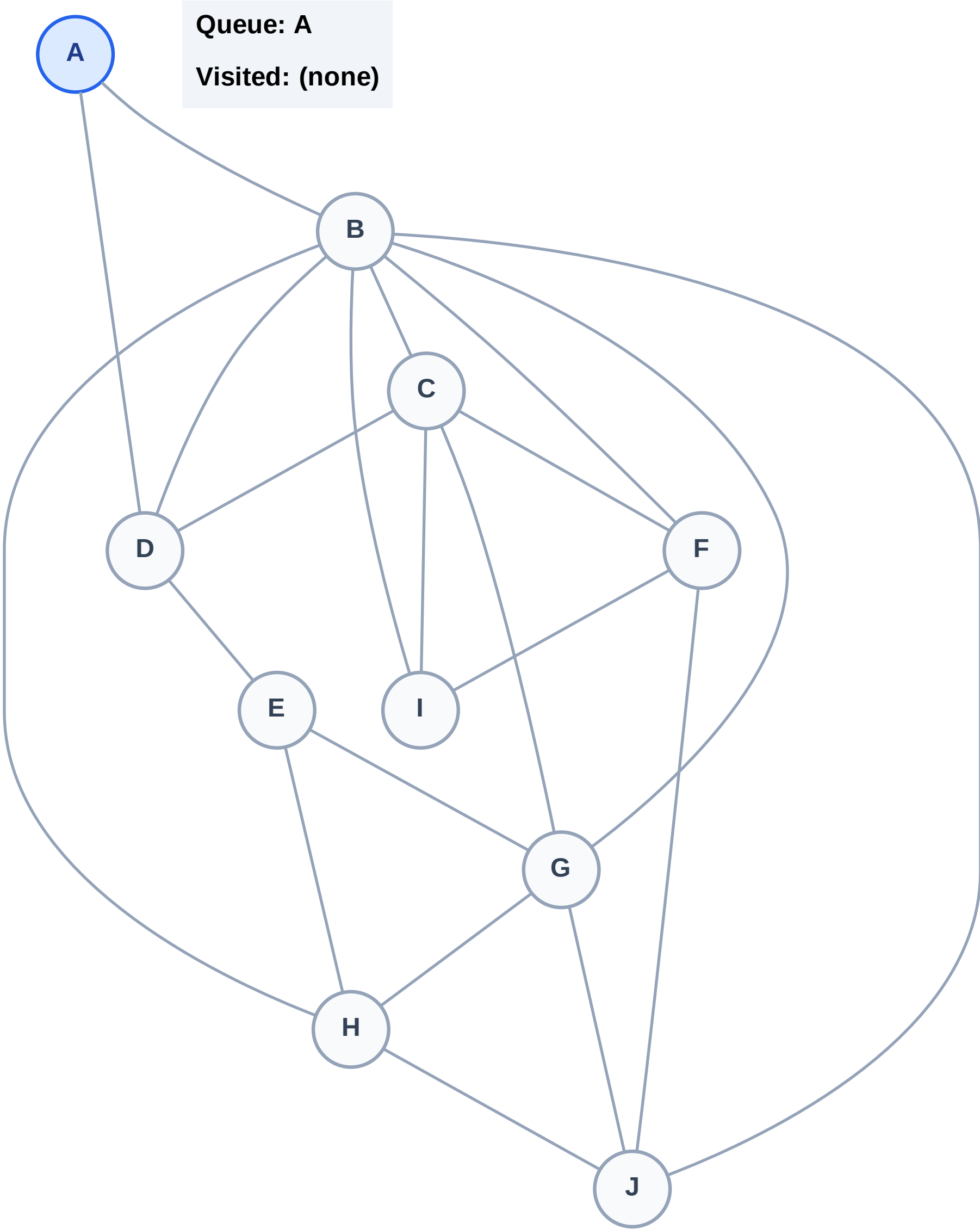
Legend

Complete

Processing

Discovered

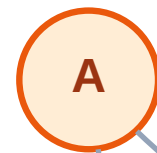
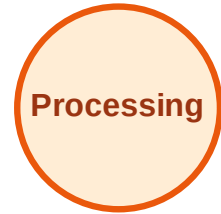
Unseen



BFS Traversal

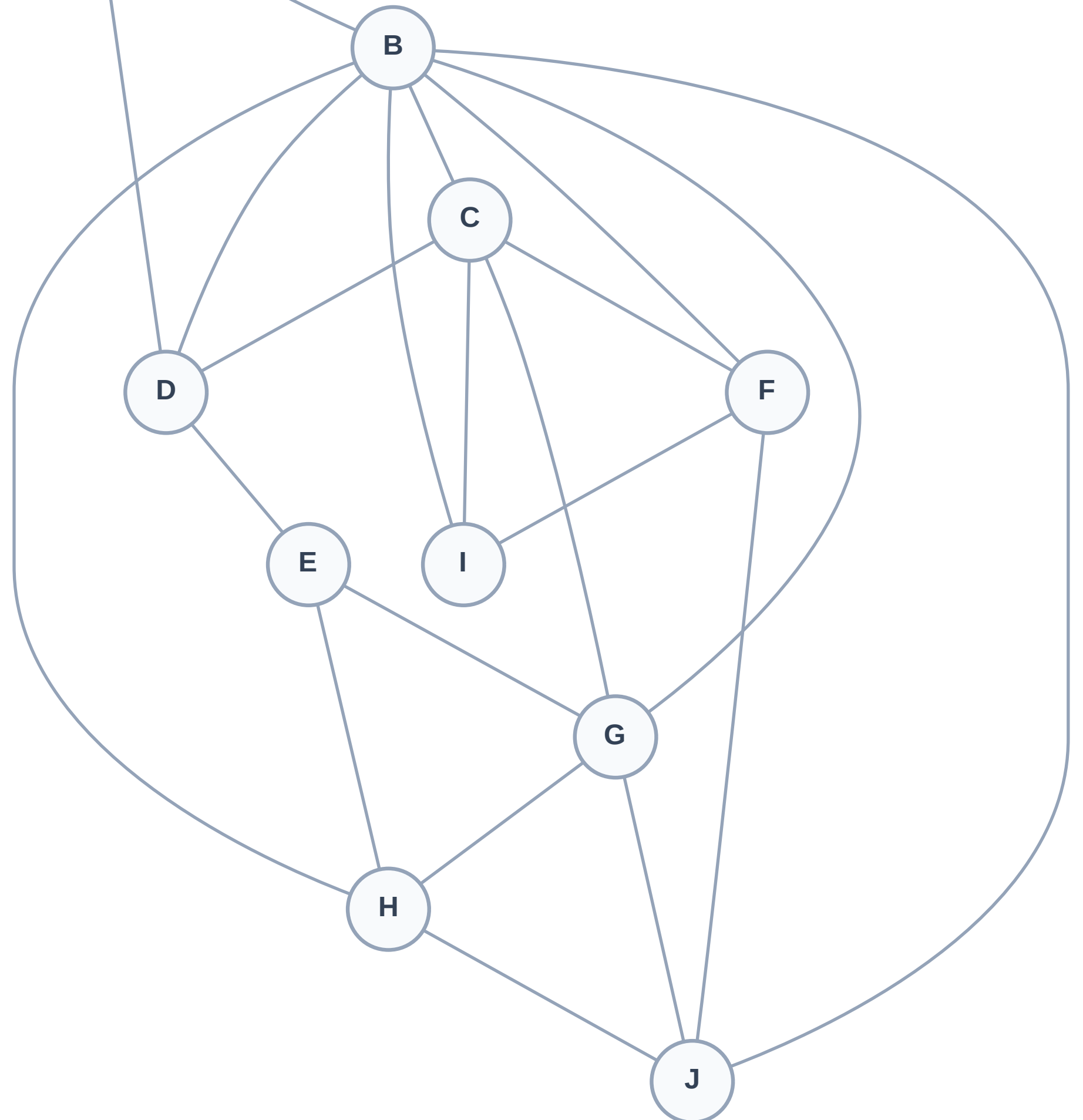
Step 2: Process node A

Legend



Queue: (empty)

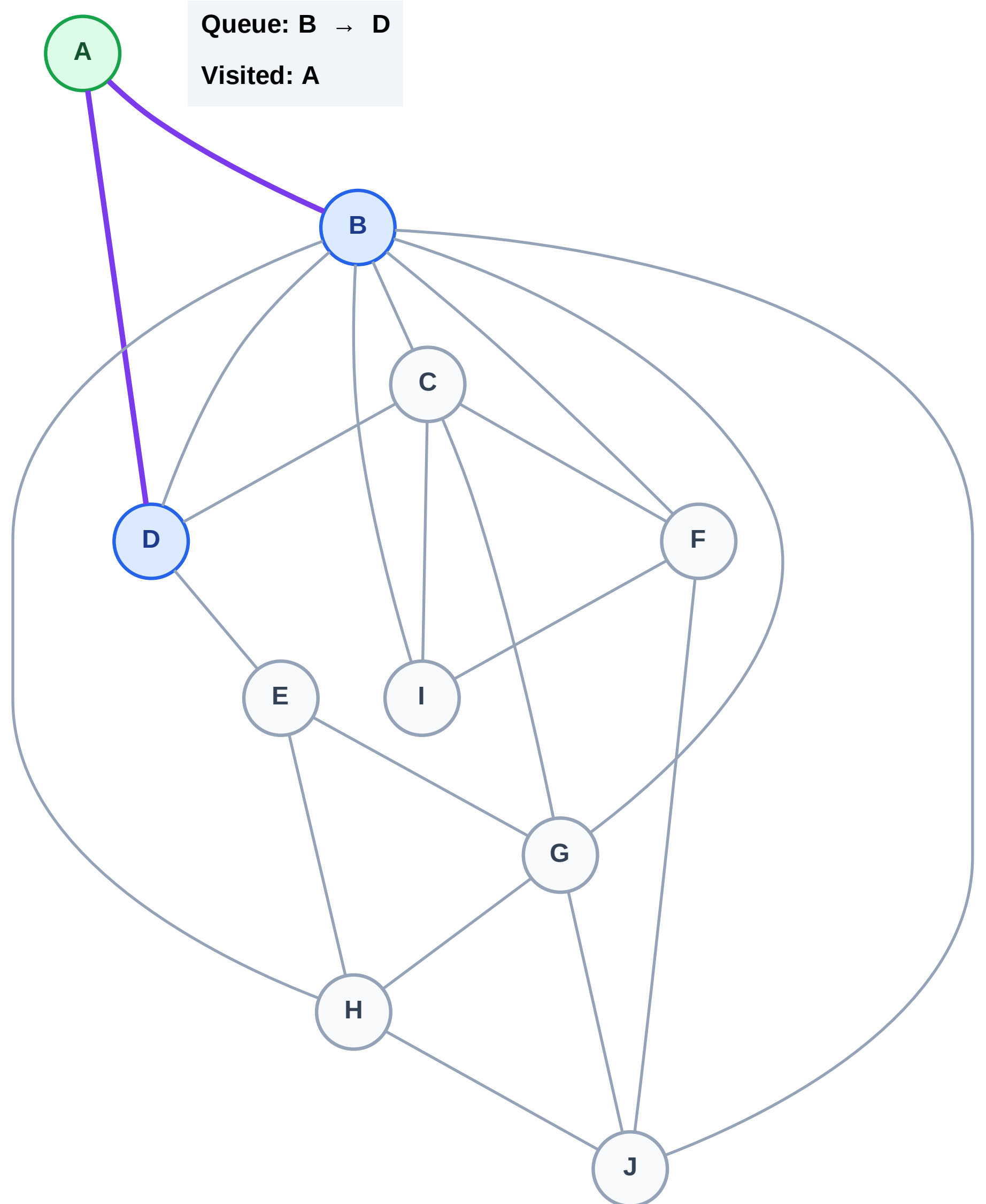
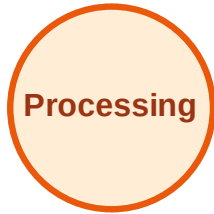
Visited: (none)



BFS Traversal

Step 3: Discover B, D from A

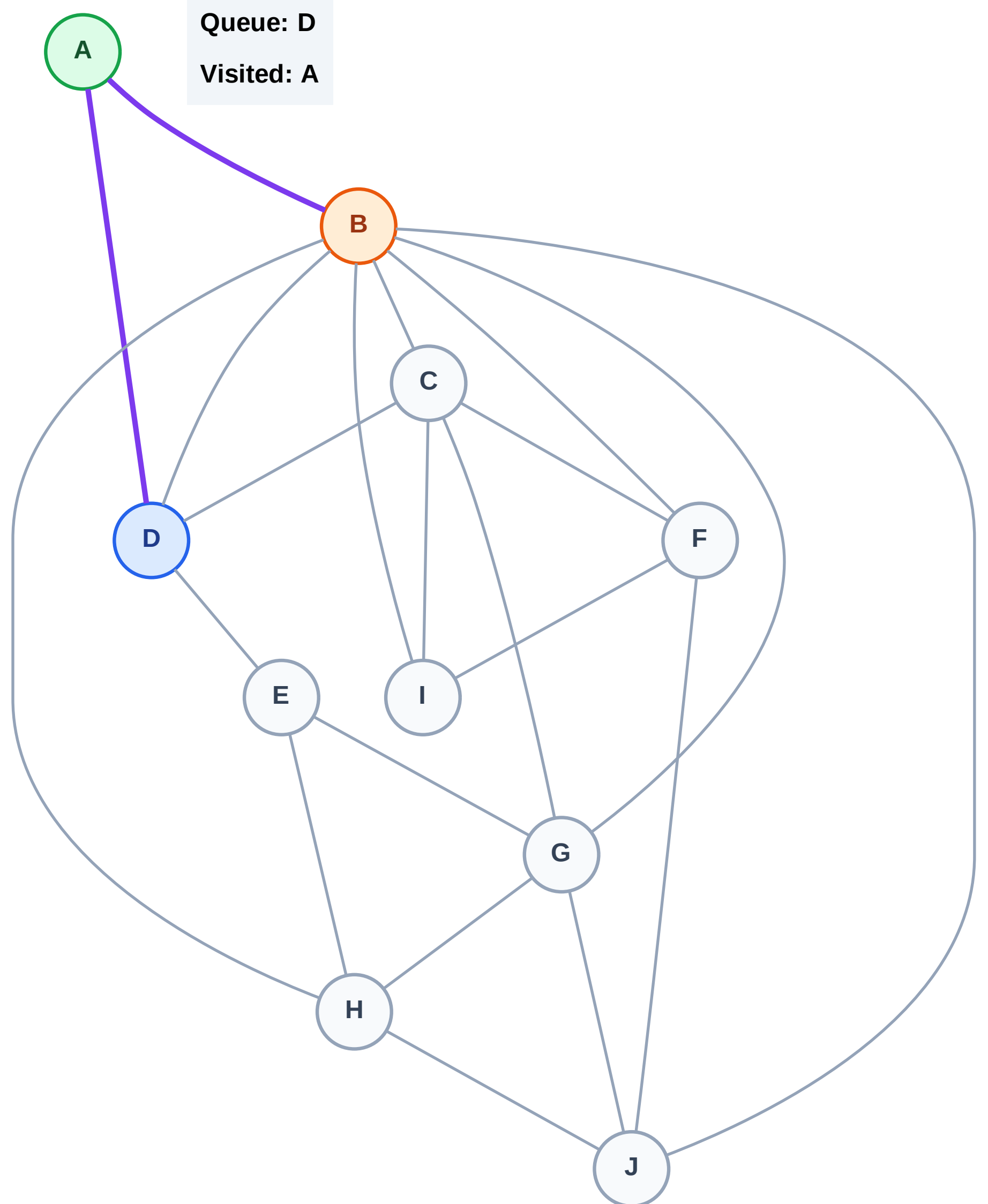
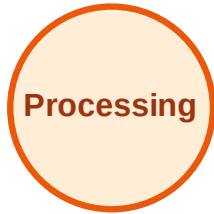
Legend



BFS Traversal

Step 4: Process node B

Legend



BFS Traversal

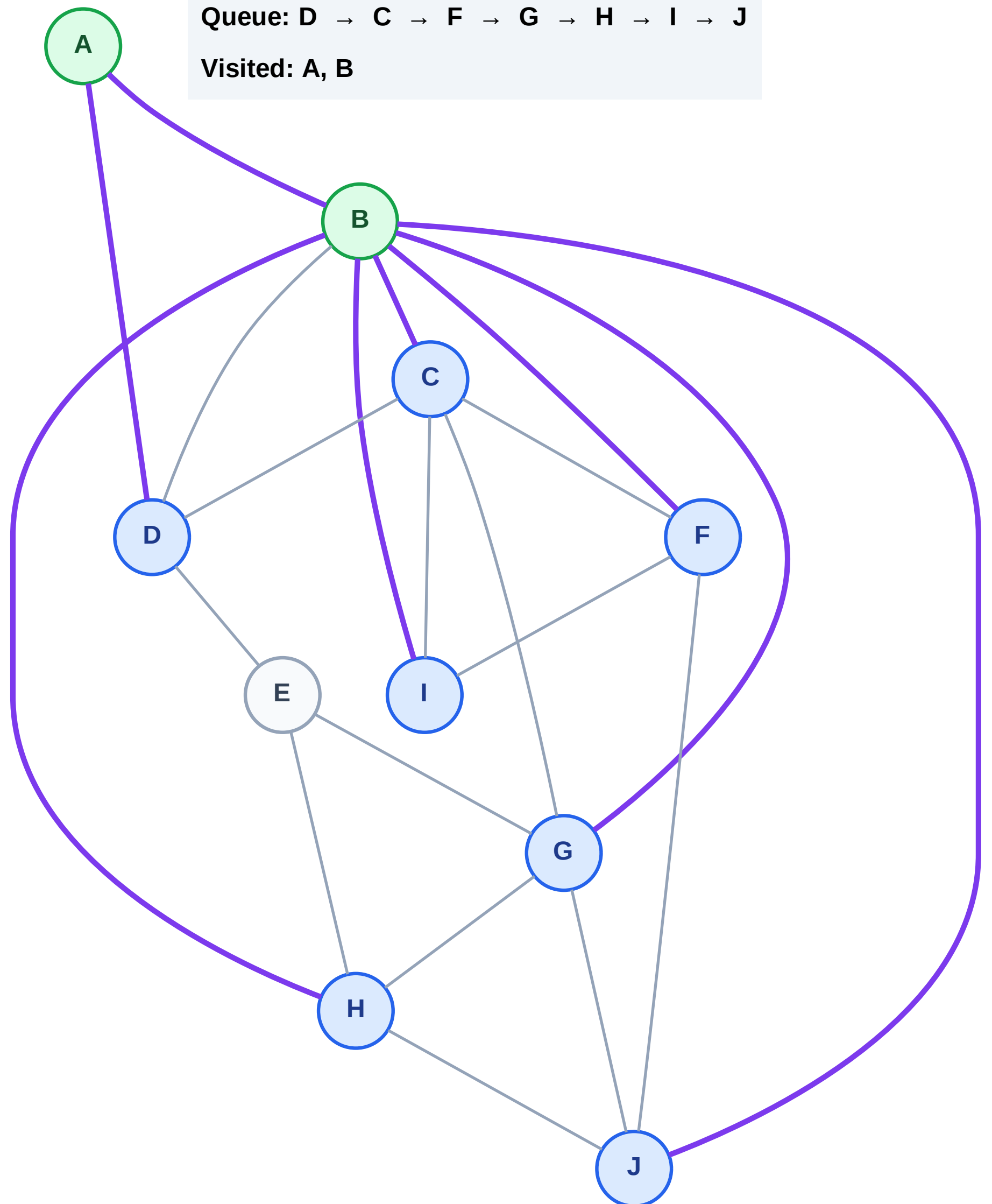
Step 5: Discover C, F, G, H, I, J from B

Legend



Queue: D → C → F → G → H → I → J

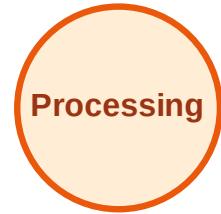
Visited: A, B



BFS Traversal

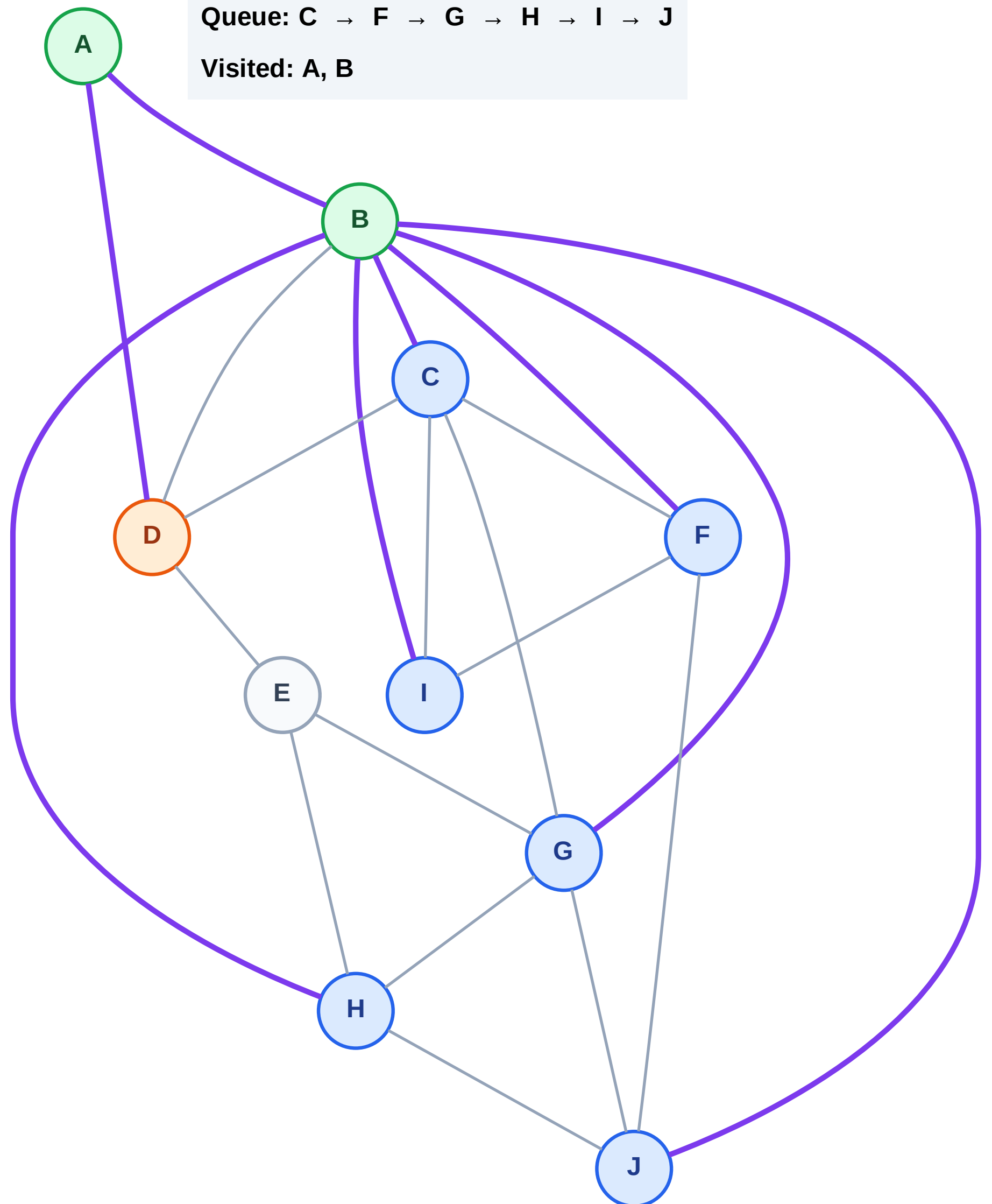
Step 6: Process node D

Legend



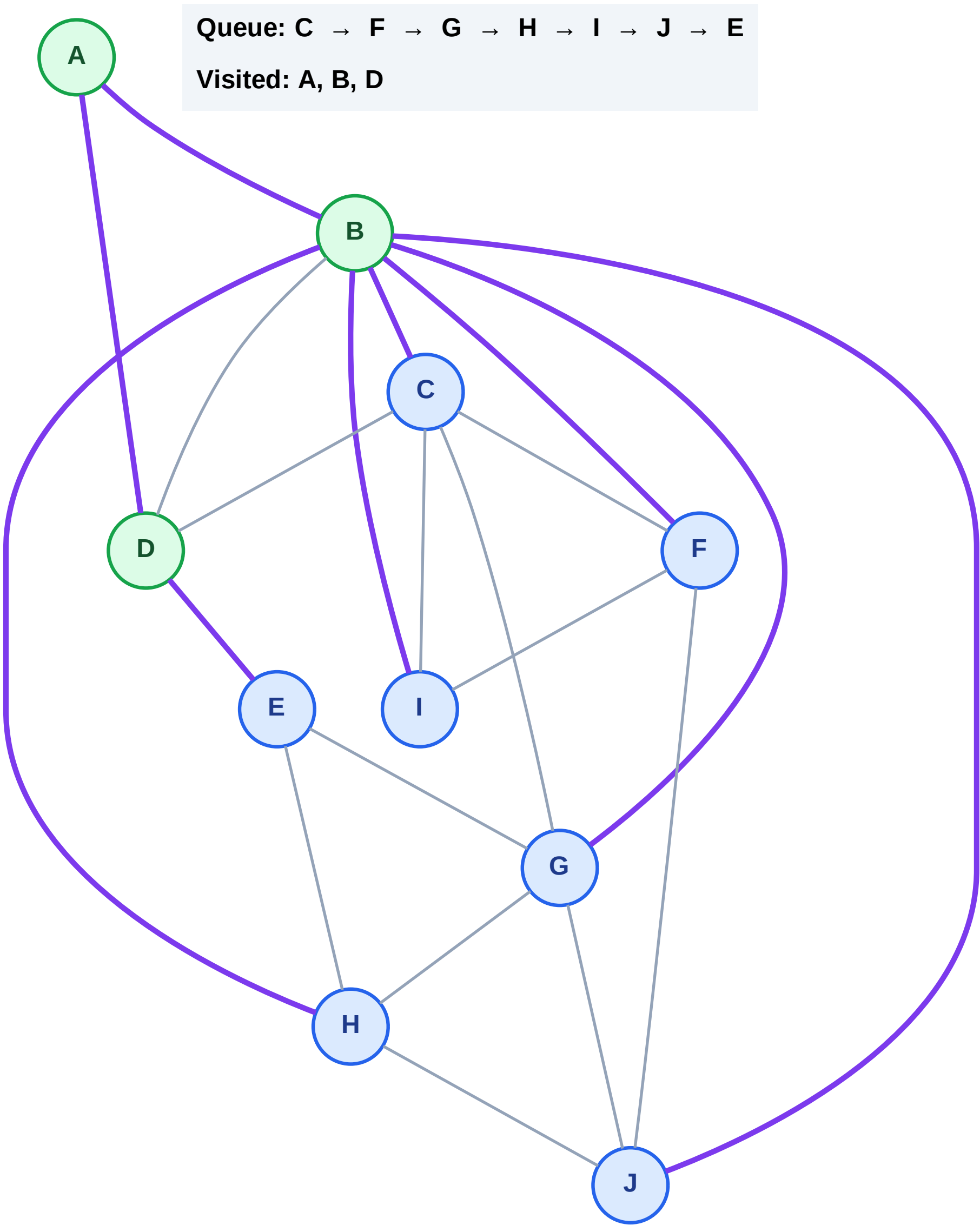
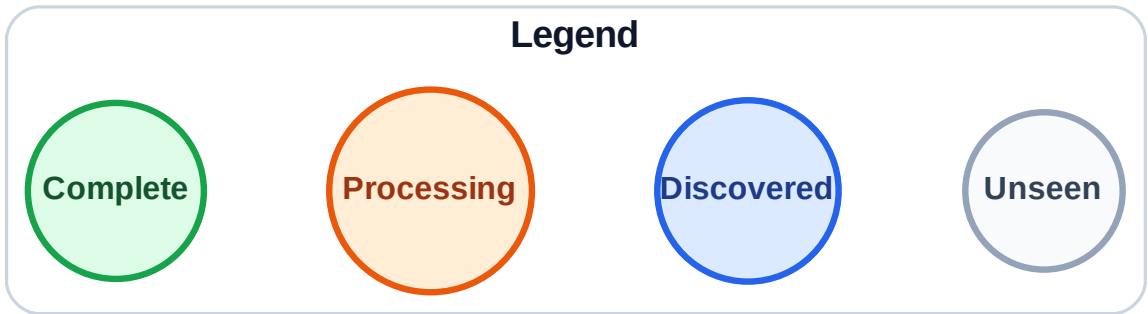
Queue: C → F → G → H → I → J

Visited: A, B

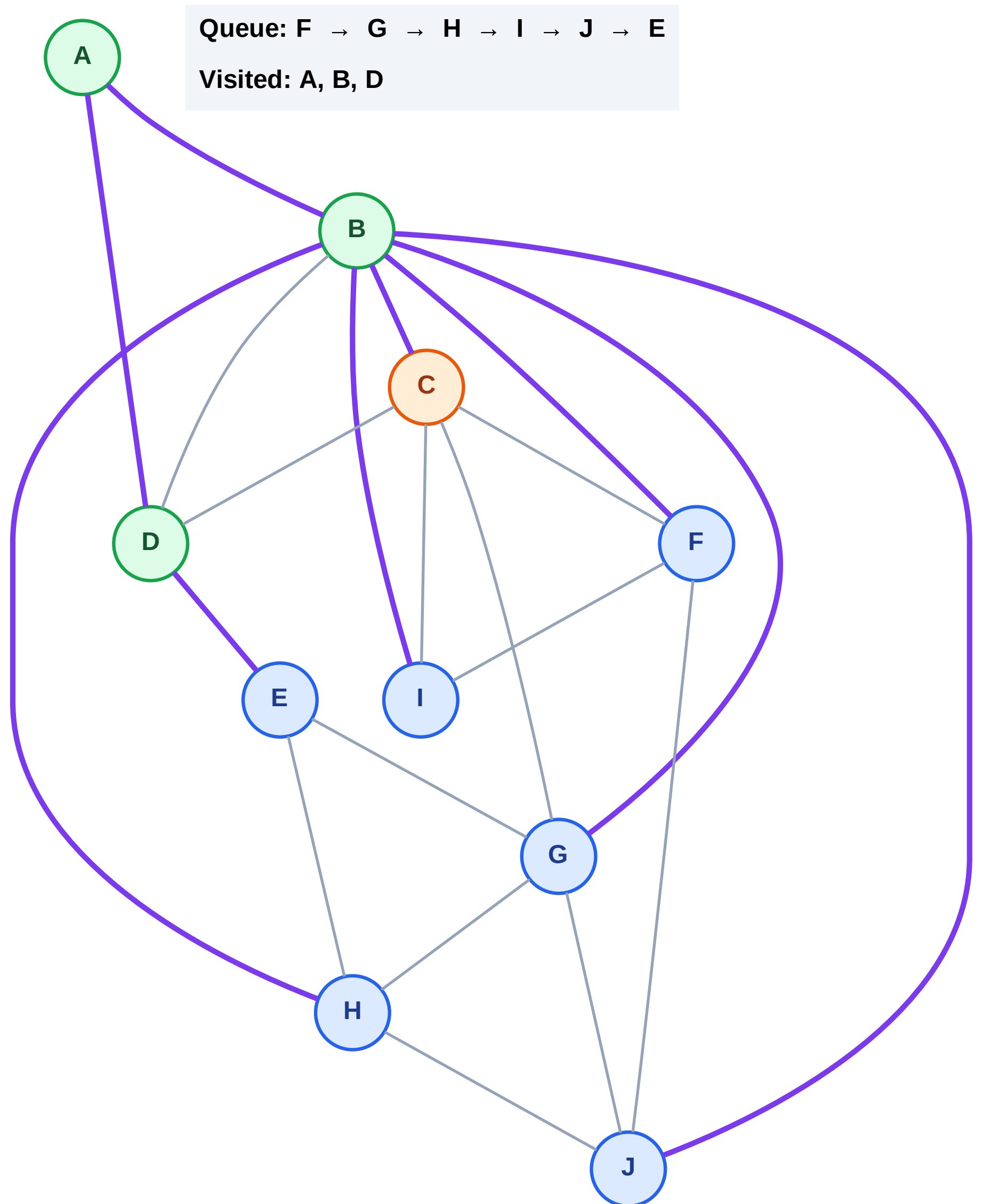
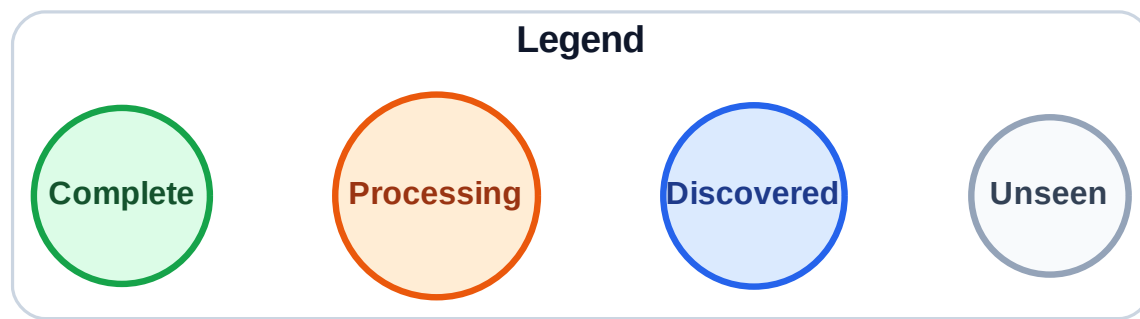


BFS Traversal

Step 7: Discover E from D



Step 8: Process node C



BFS Traversal

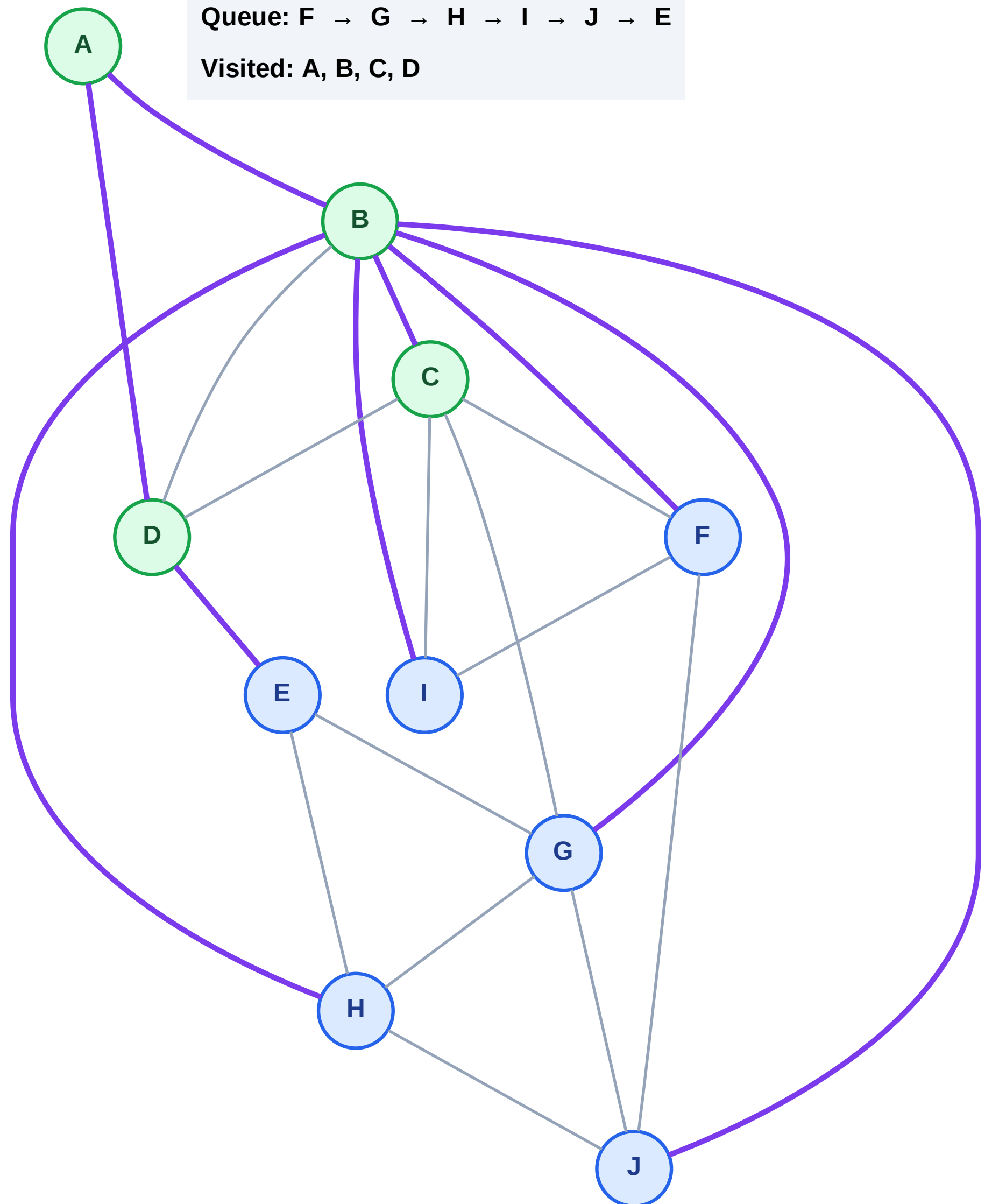
Step 9: No new neighbors from C

Legend



Queue: F → G → H → I → J → E

Visited: A, B, C, D

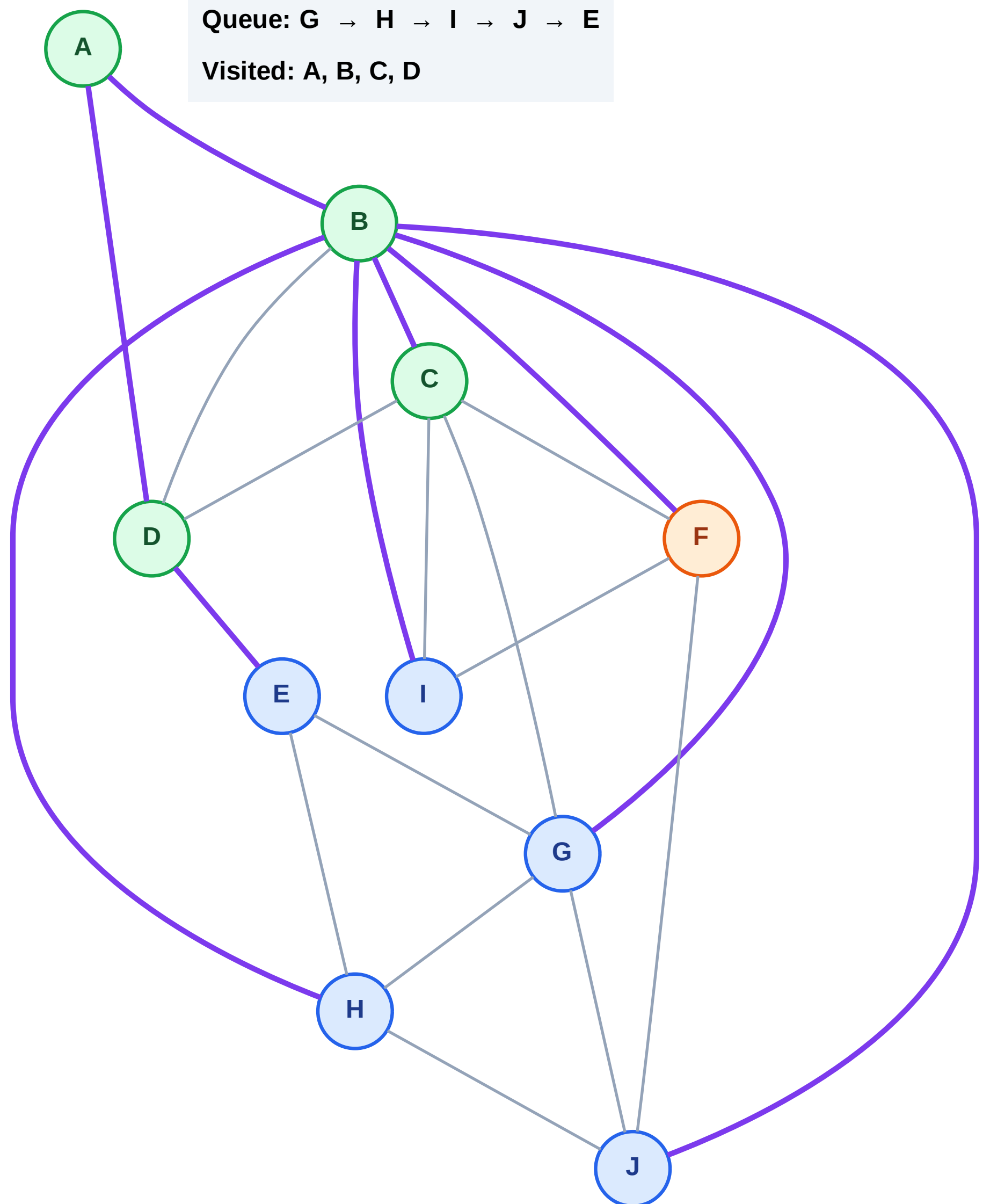
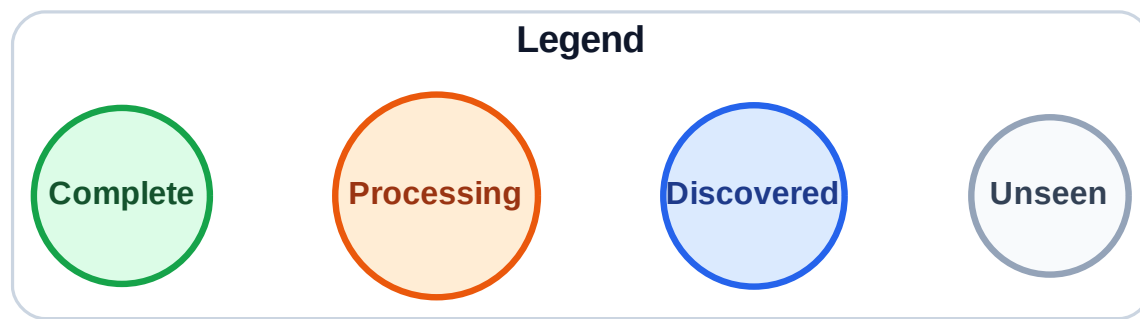


BFS Traversal

Step 10: Process node F

Queue: G → H → I → J → E

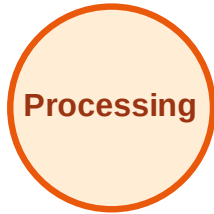
Visited: A, B, C, D



BFS Traversal

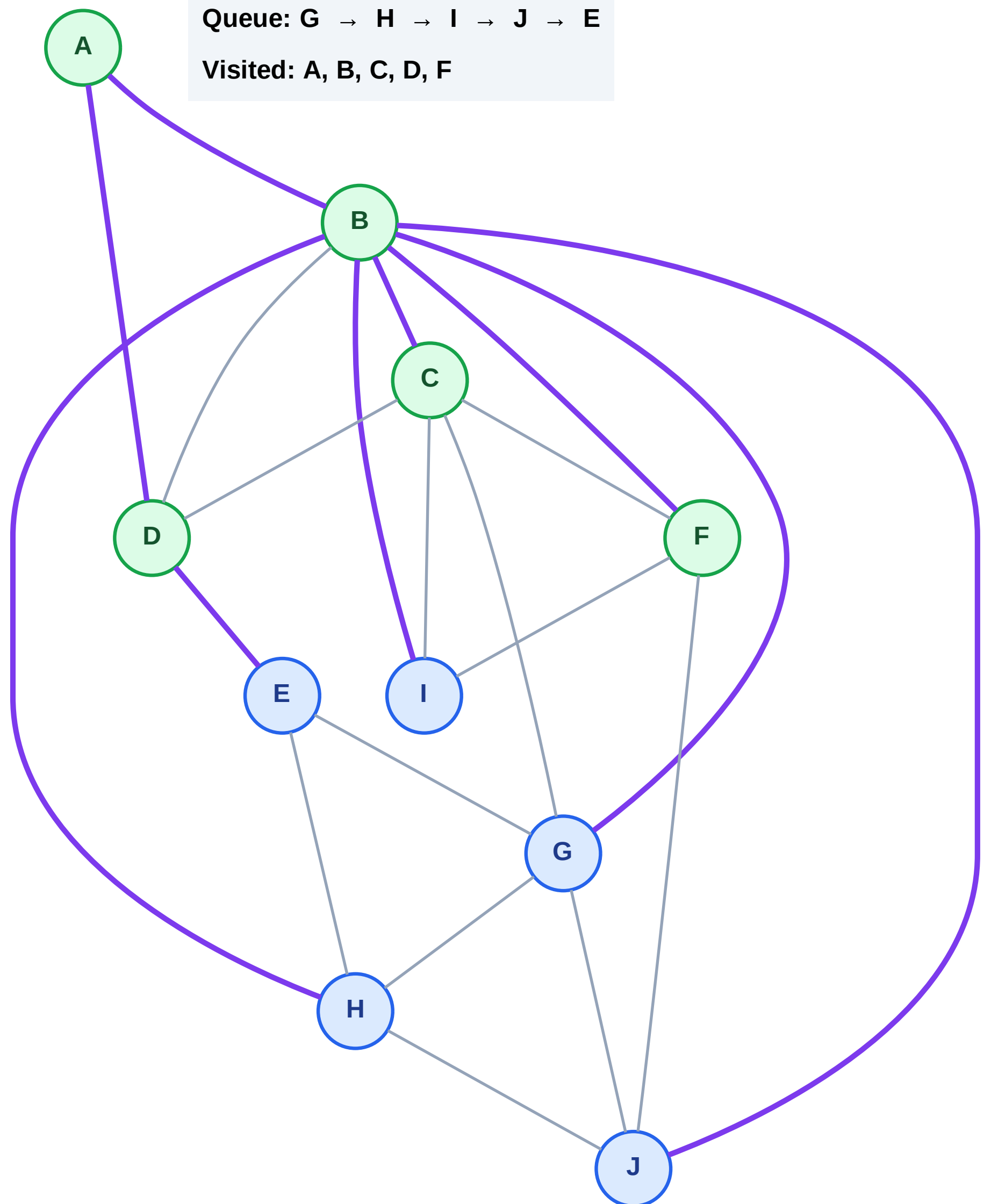
Step 11: No new neighbors from F

Legend



Queue: G → H → I → J → E

Visited: A, B, C, D, F



BFS Traversal

Step 12: Process node G

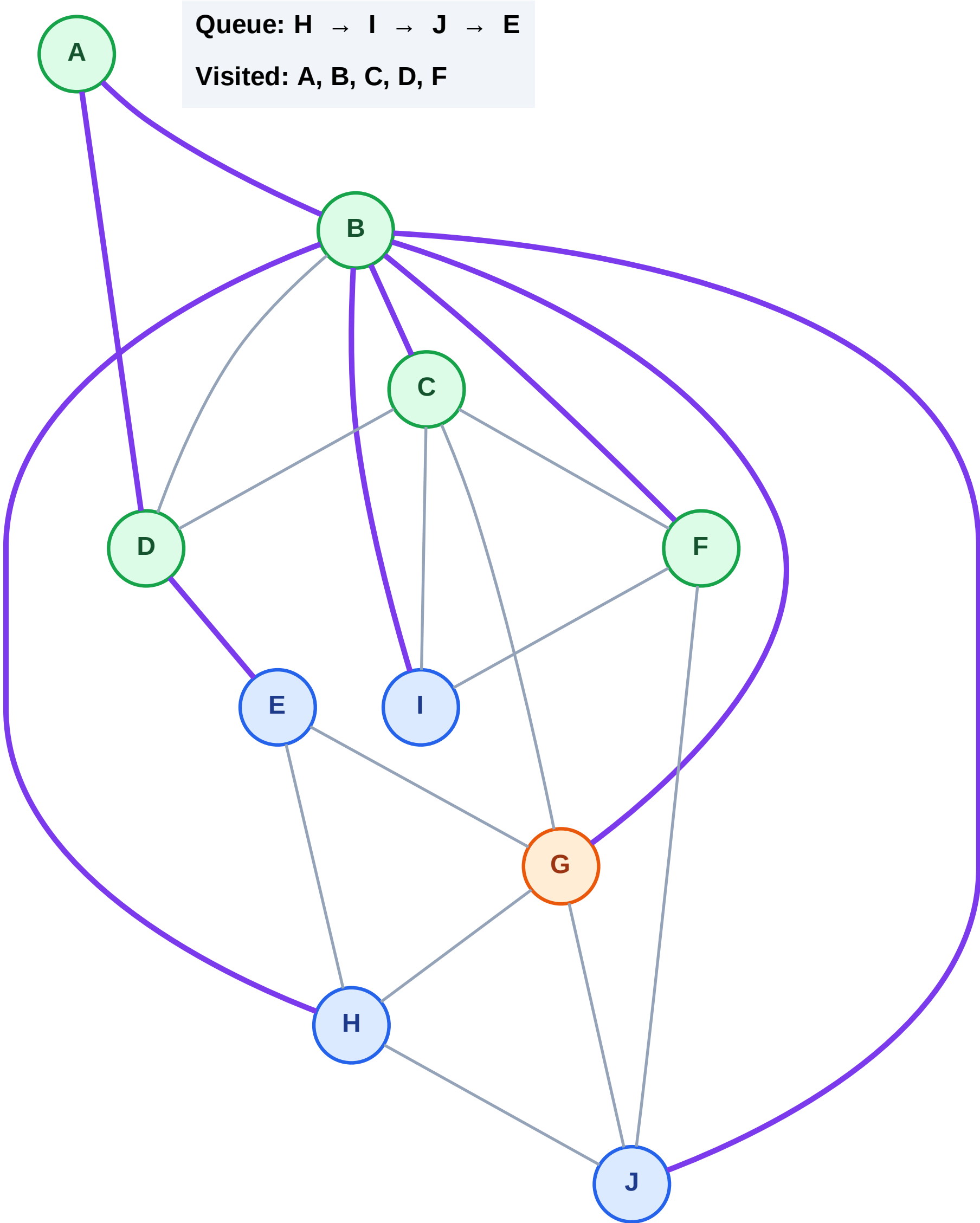
Legend

Complete

Processing

Discovered

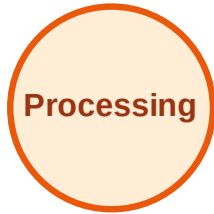
Unseen



BFS Traversal

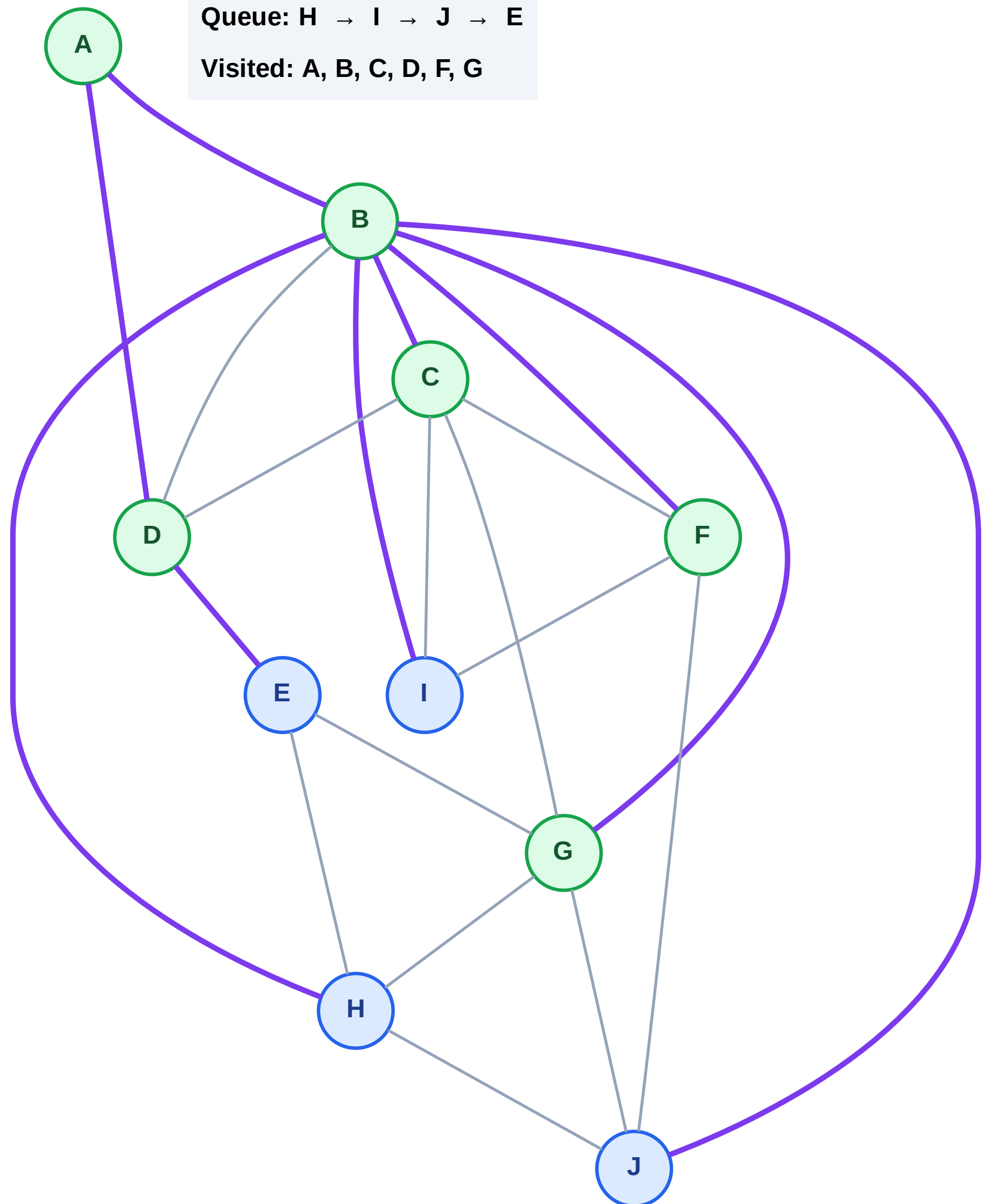
Step 13: No new neighbors from G

Legend



Queue: H → I → J → E

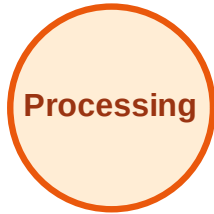
Visited: A, B, C, D, F, G



BFS Traversal

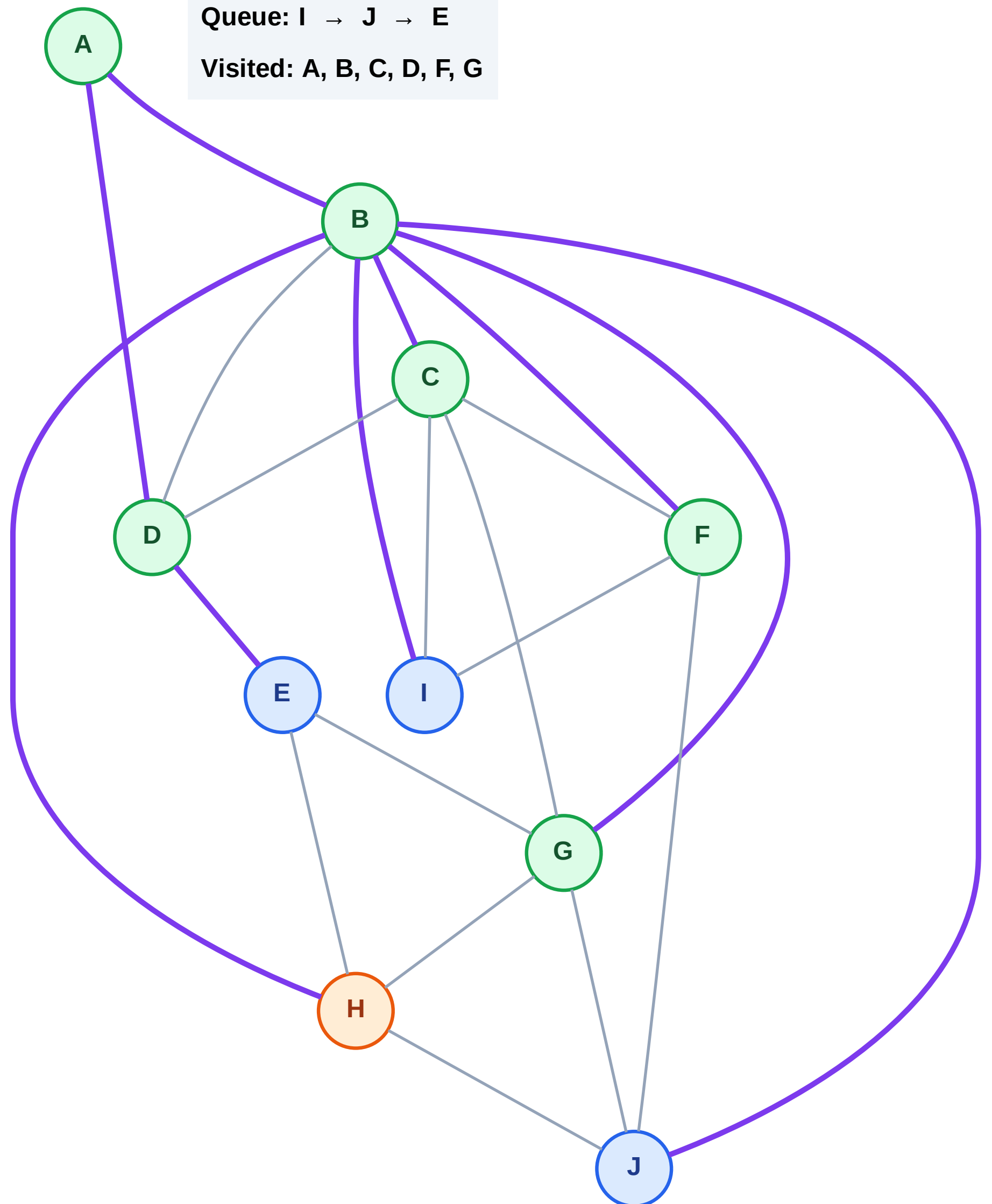
Step 14: Process node H

Legend



Queue: I → J → E

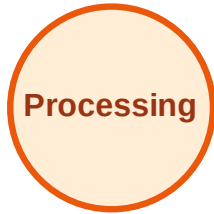
Visited: A, B, C, D, F, G



BFS Traversal

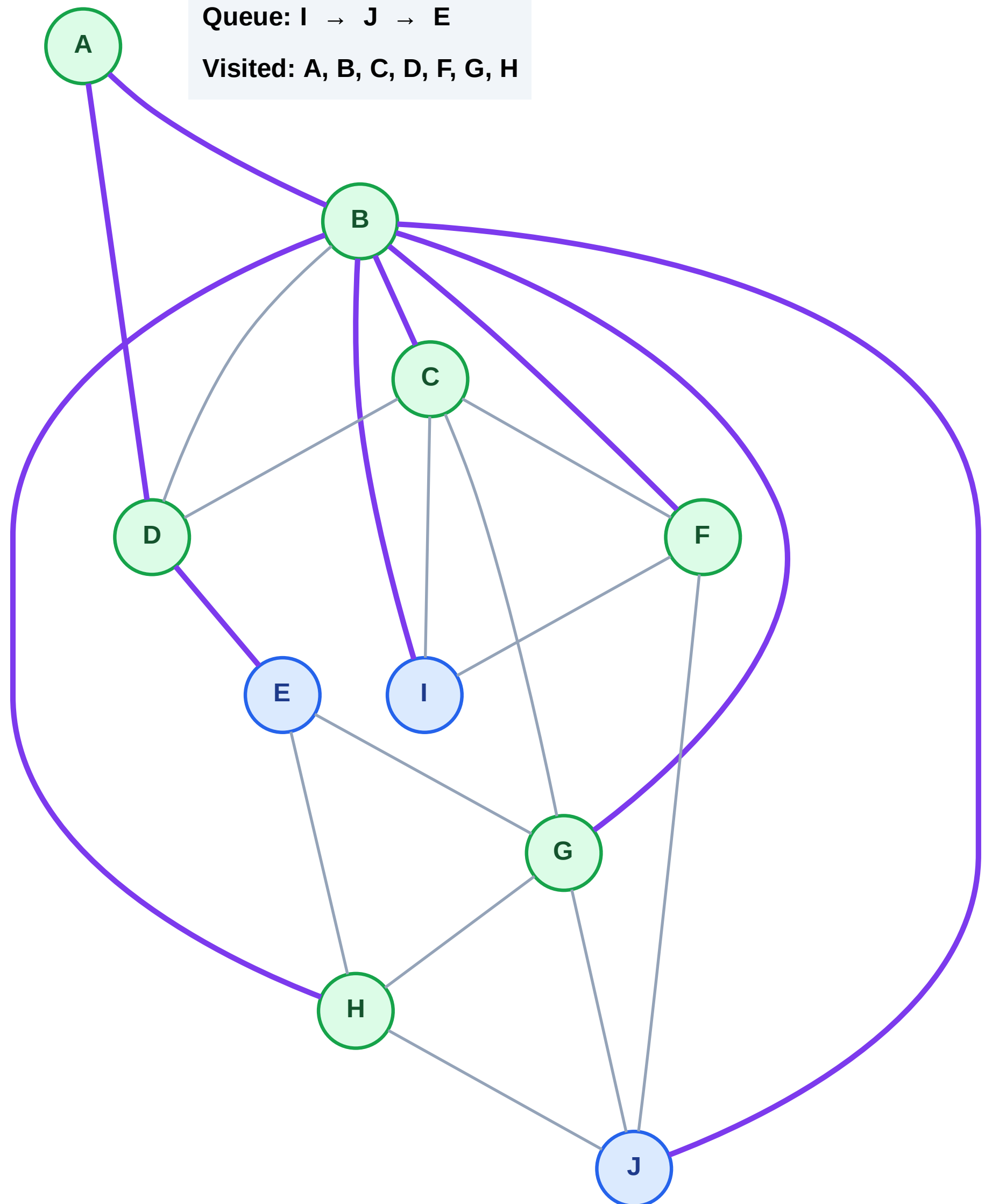
Step 15: No new neighbors from H

Legend

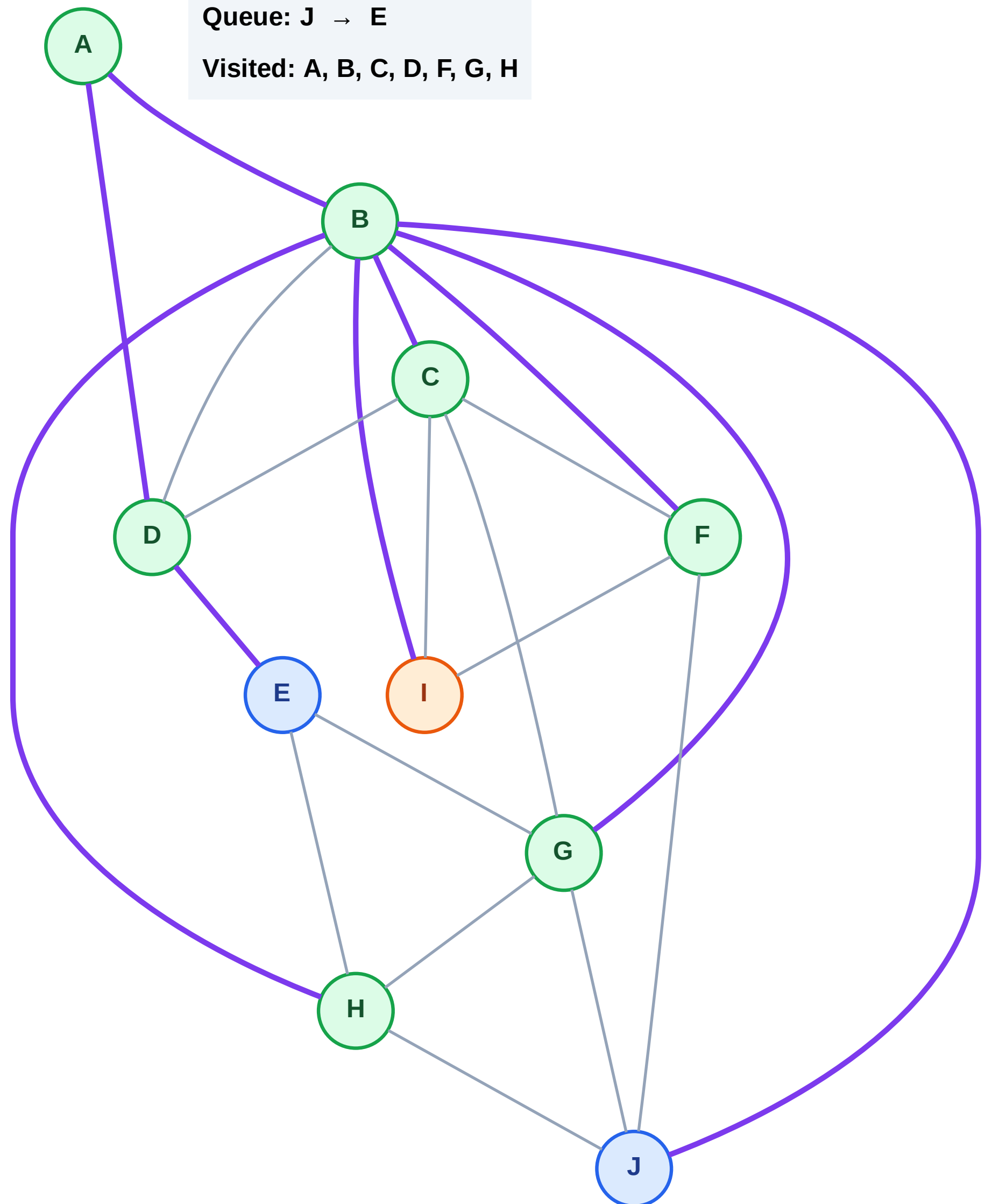
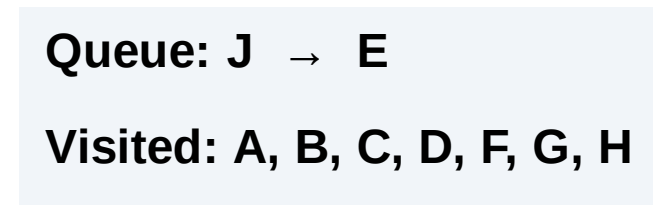


Queue: I → J → E

Visited: A, B, C, D, F, G, H



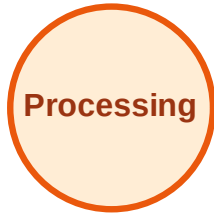
Step 16: Process node I



BFS Traversal

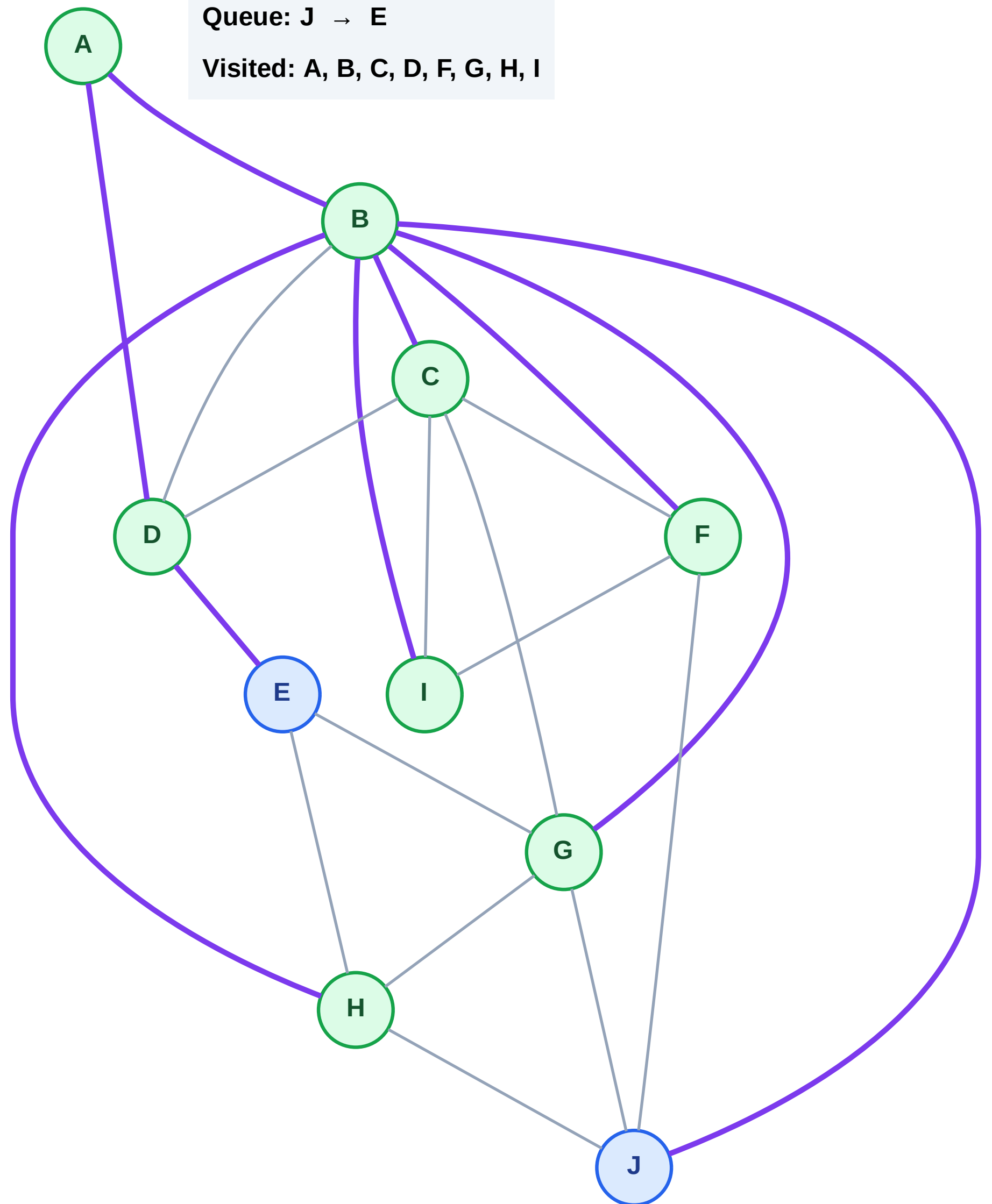
Step 17: No new neighbors from I

Legend

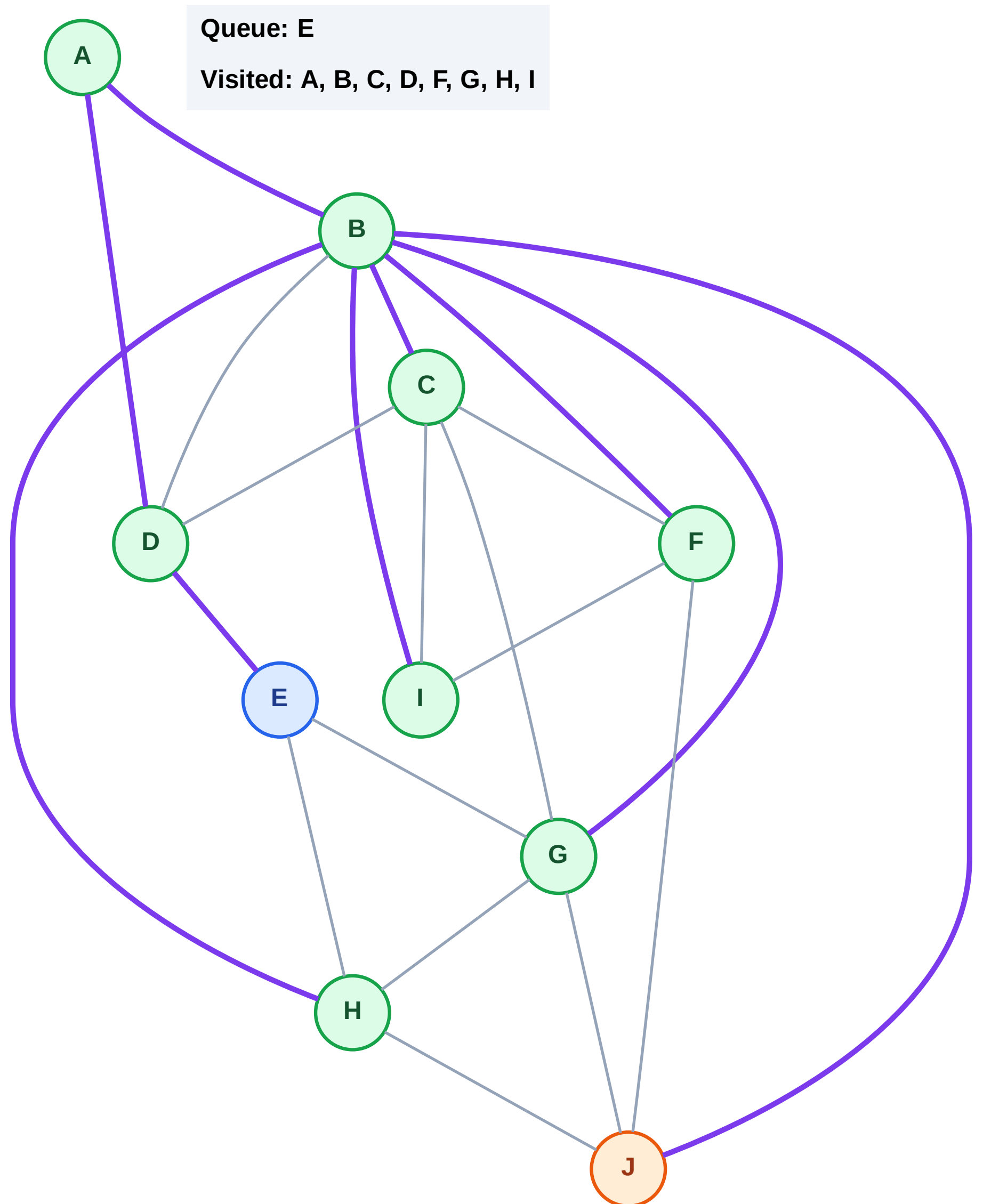
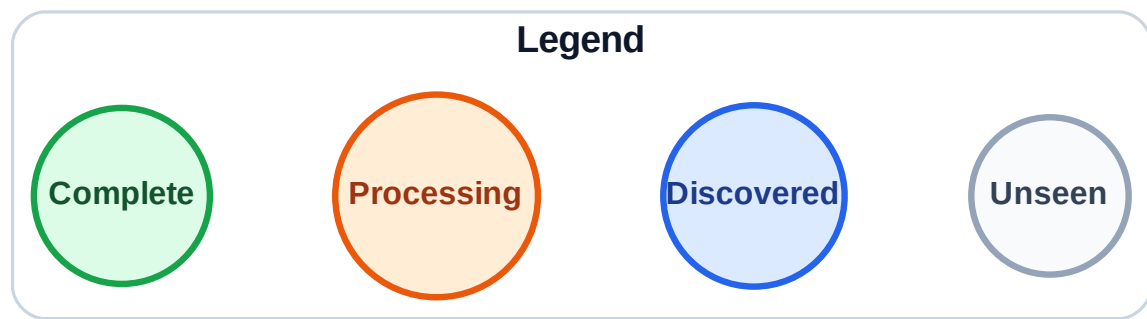


Queue: J → E

Visited: A, B, C, D, F, G, H, I



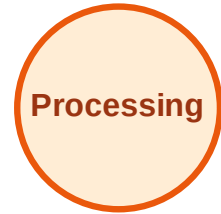
Step 18: Process node J



BFS Traversal

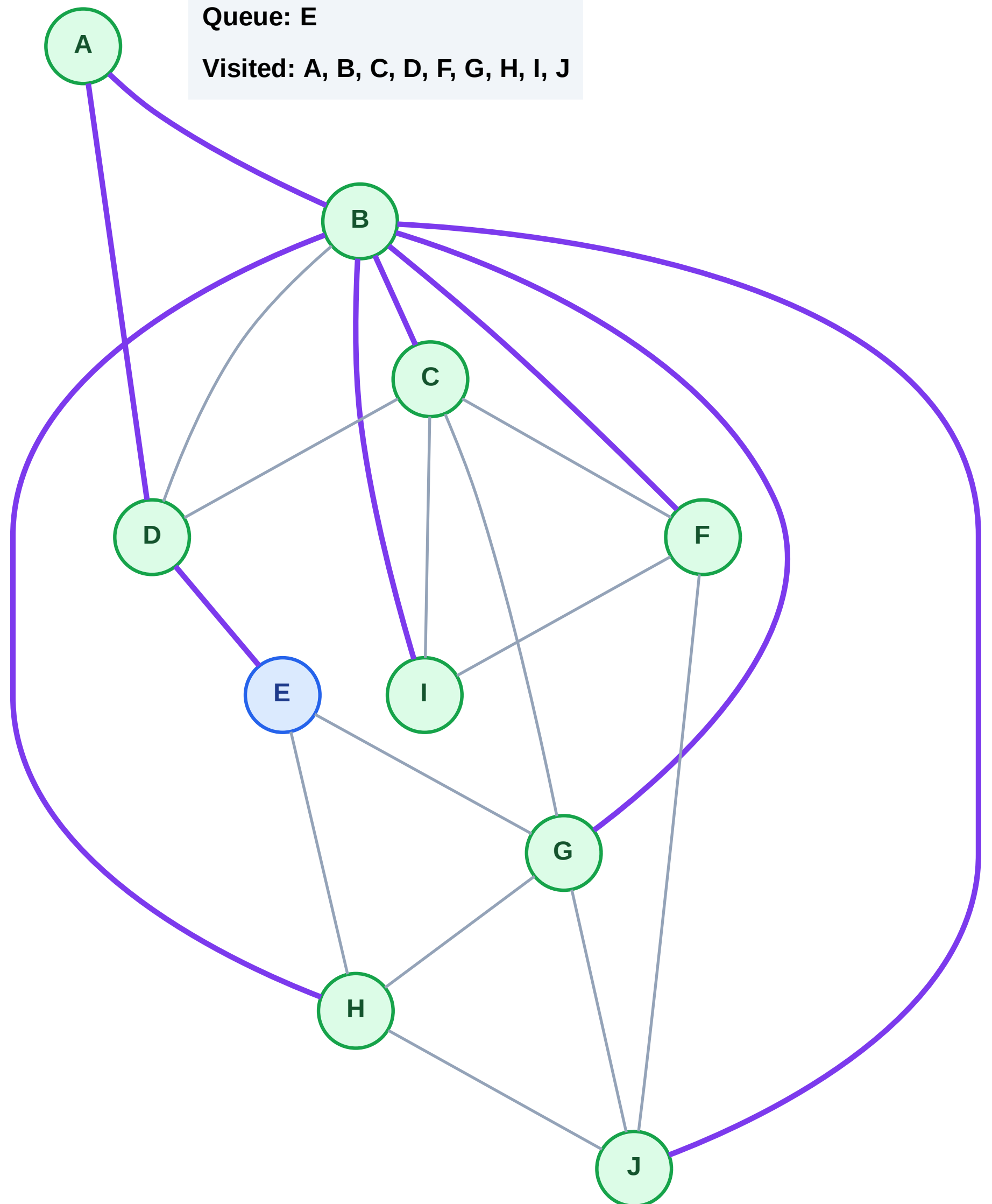
Step 19: No new neighbors from J

Legend



Queue: E

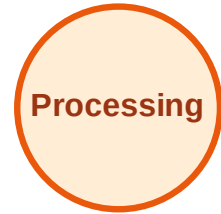
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

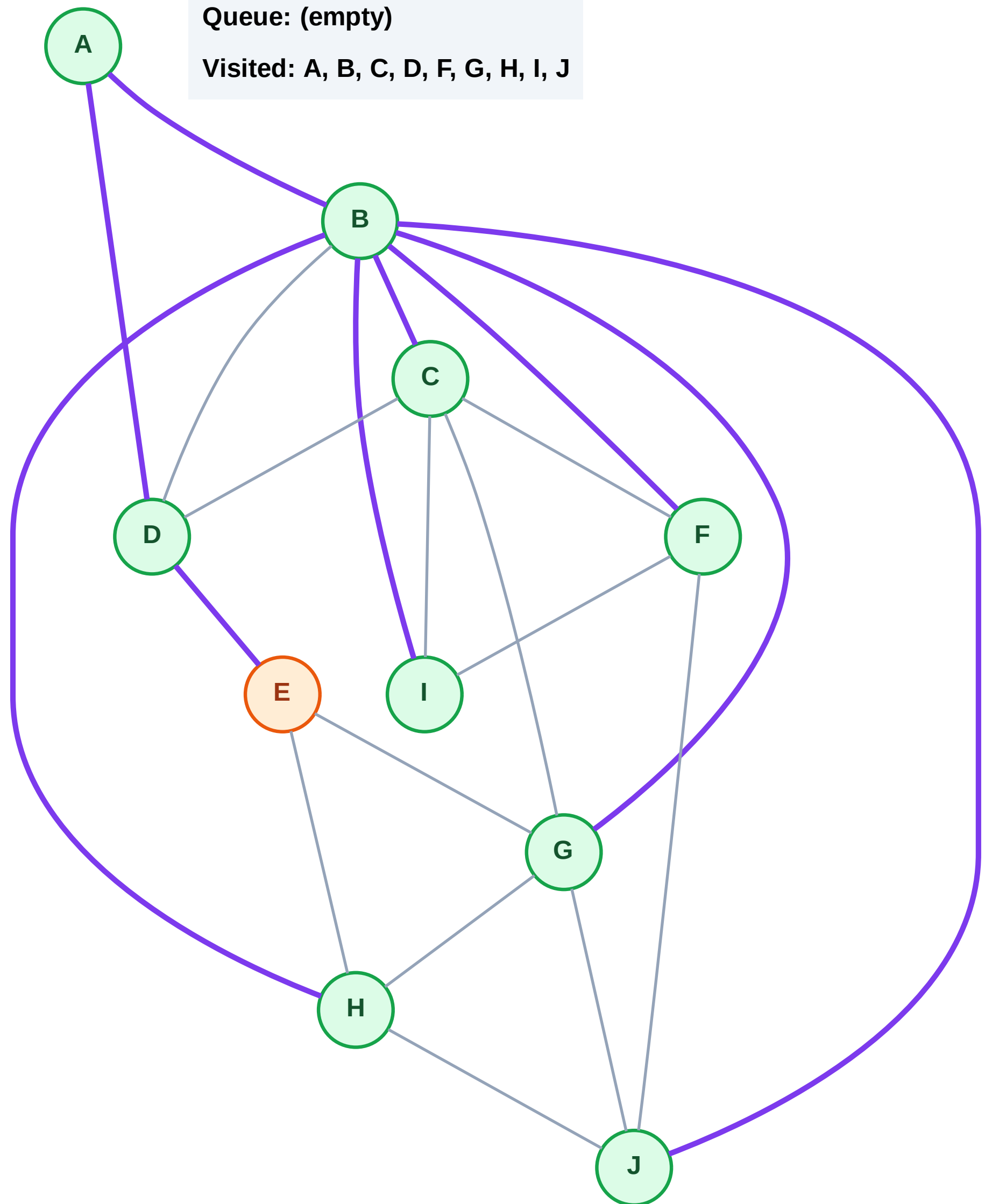
Step 20: Process node E

Legend



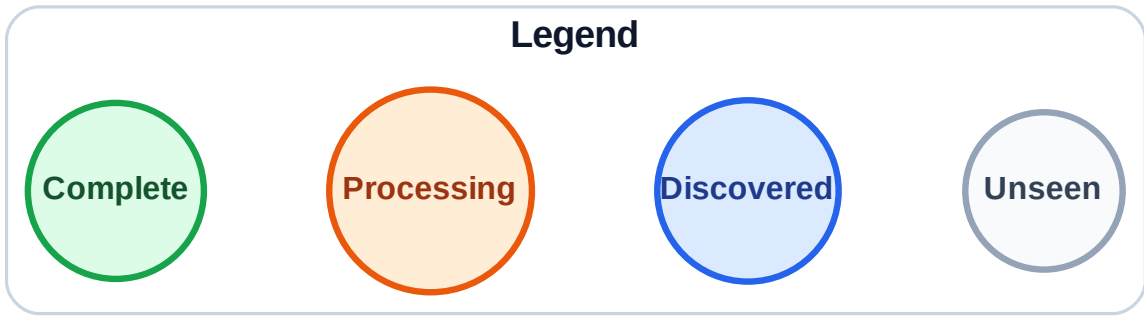
Queue: (empty)

Visited: A, B, C, D, F, G, H, I, J



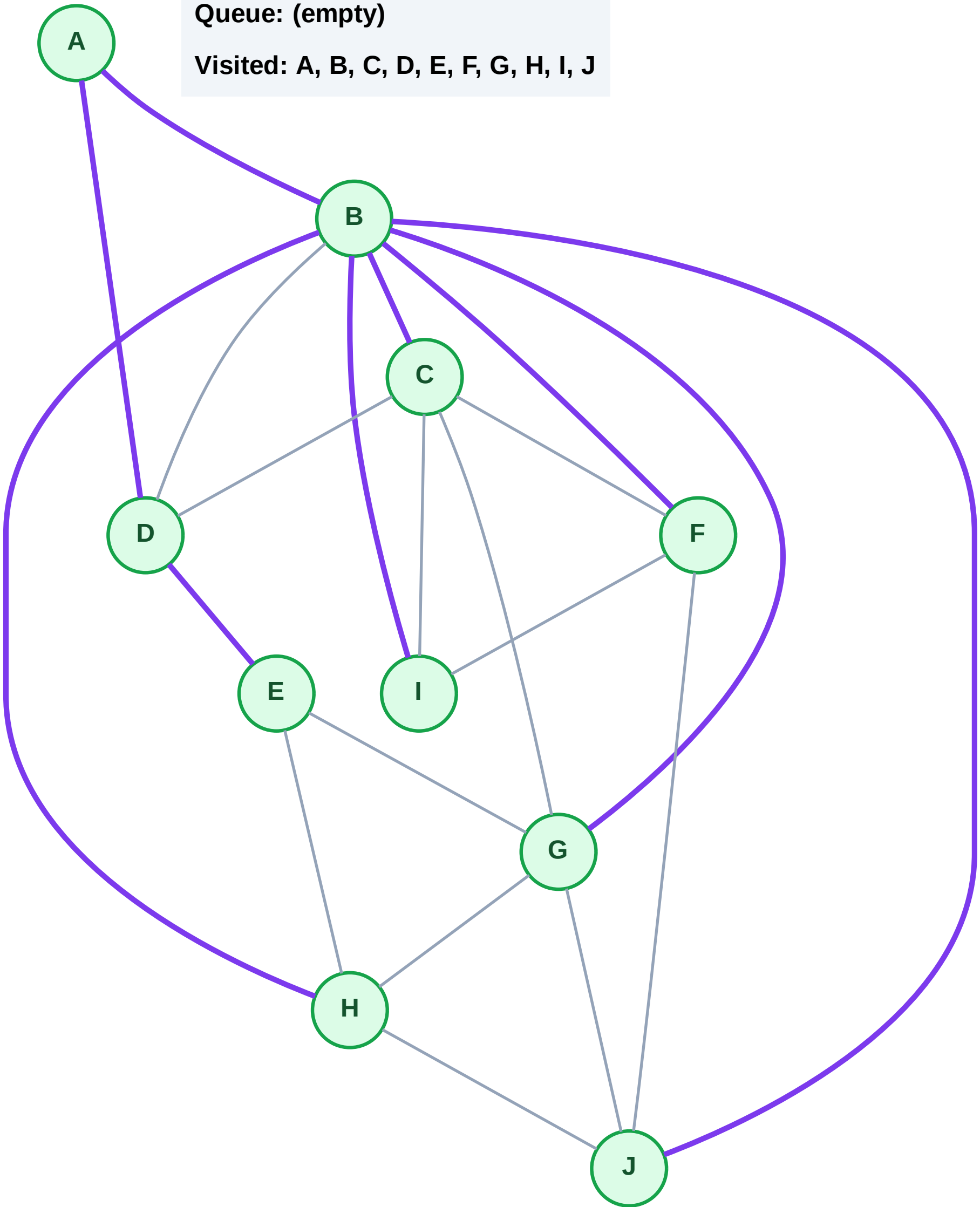
BFS Traversal

Step 21: No new neighbors from E



Queue: (empty)

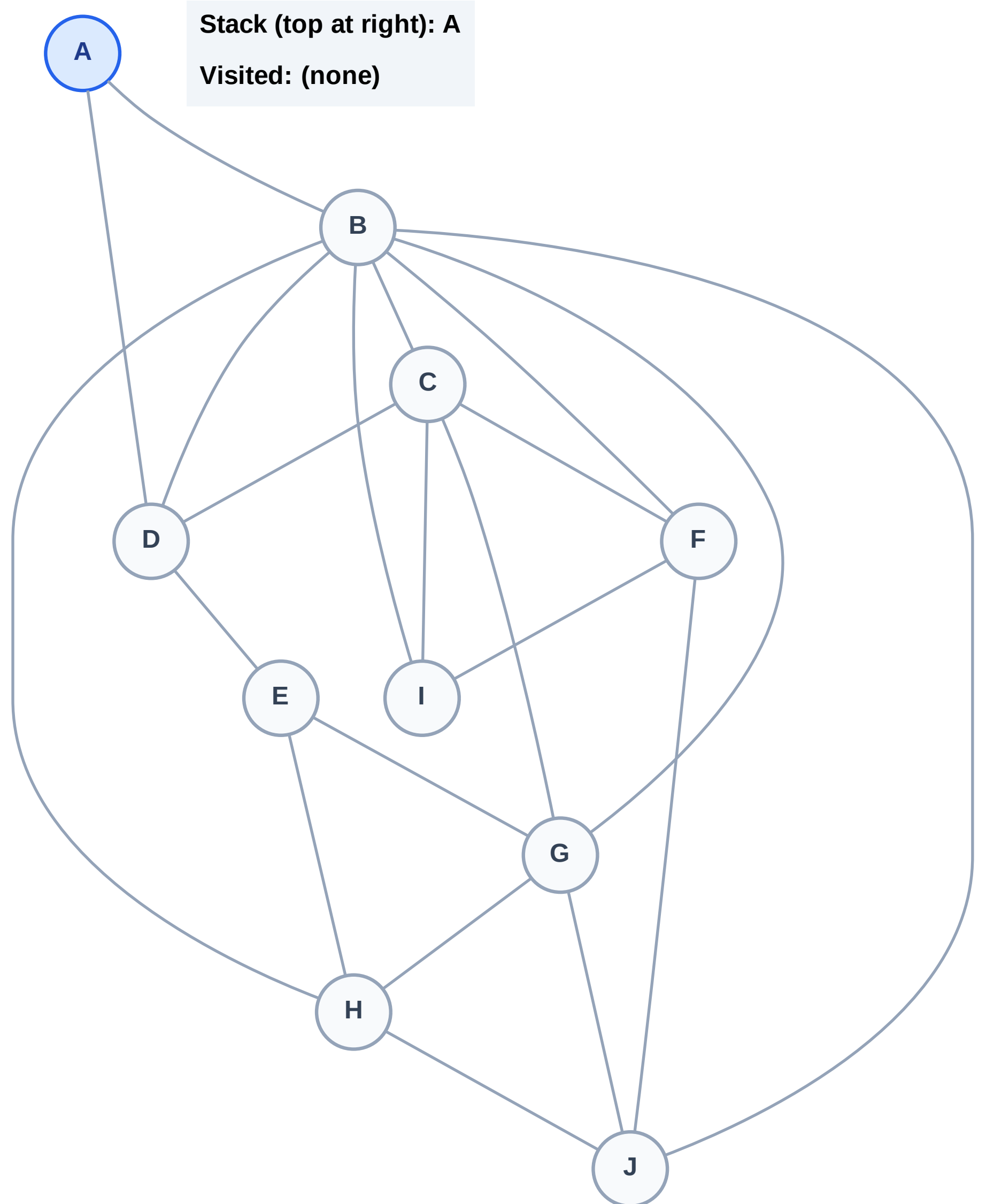
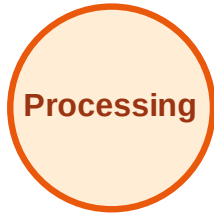
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

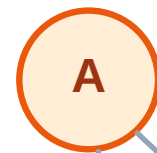
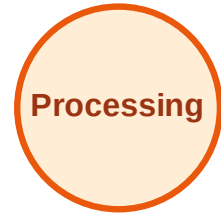
Legend



DFS Traversal

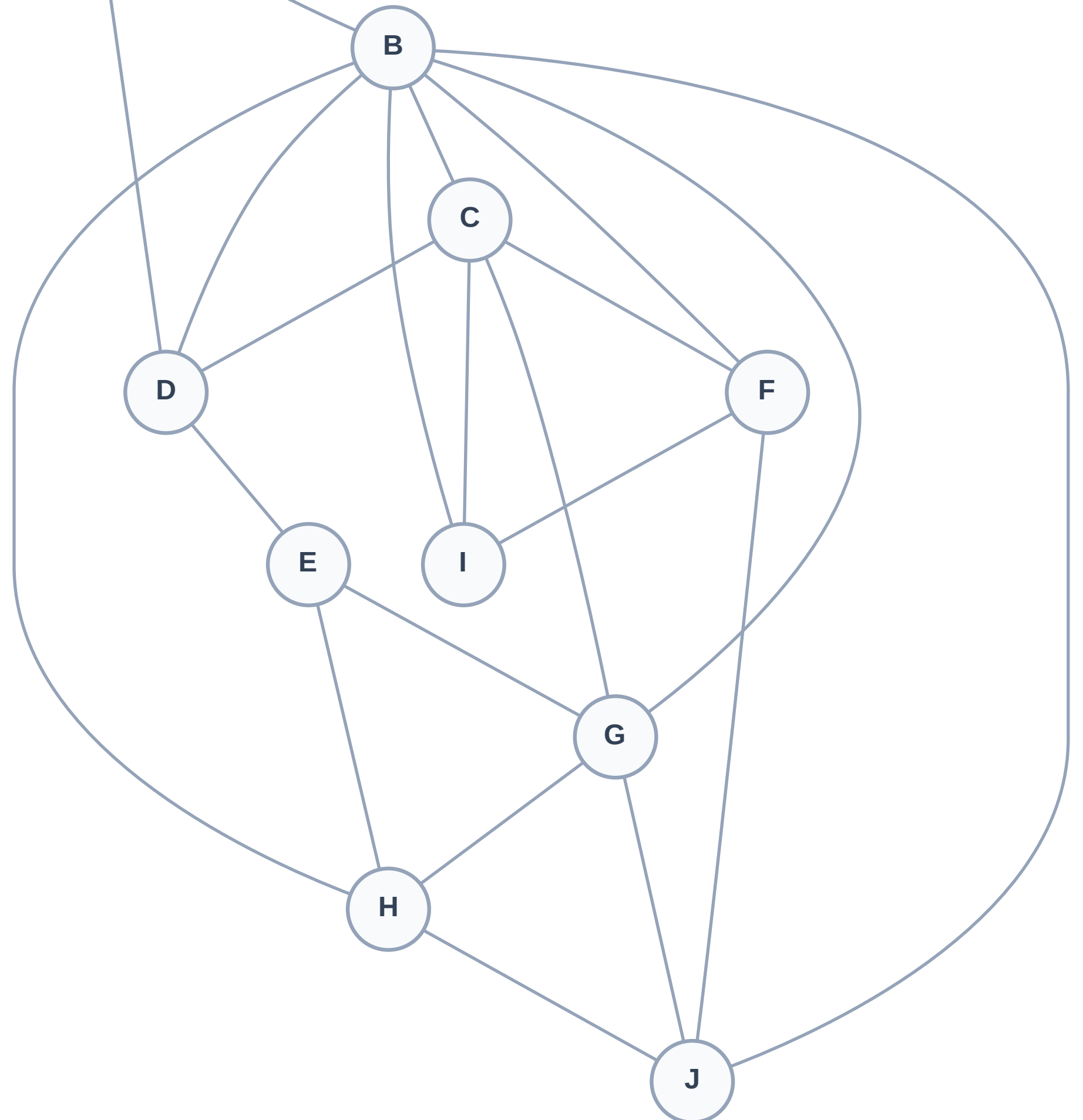
Step 2: Process node A

Legend



Stack (top at right): (empty)

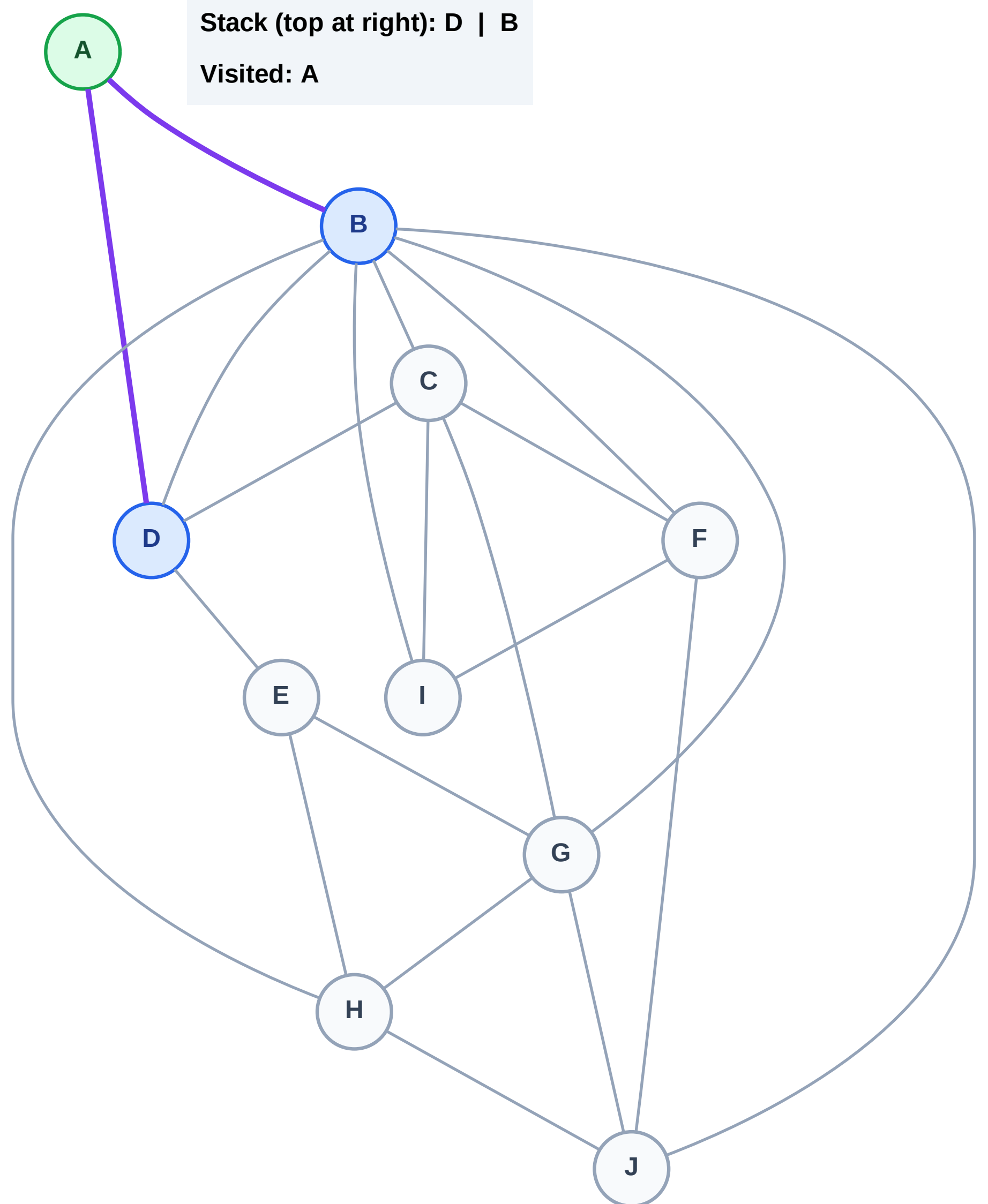
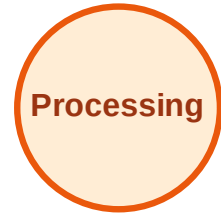
Visited: (none)



DFS Traversal

Step 3: Discover D, B from A

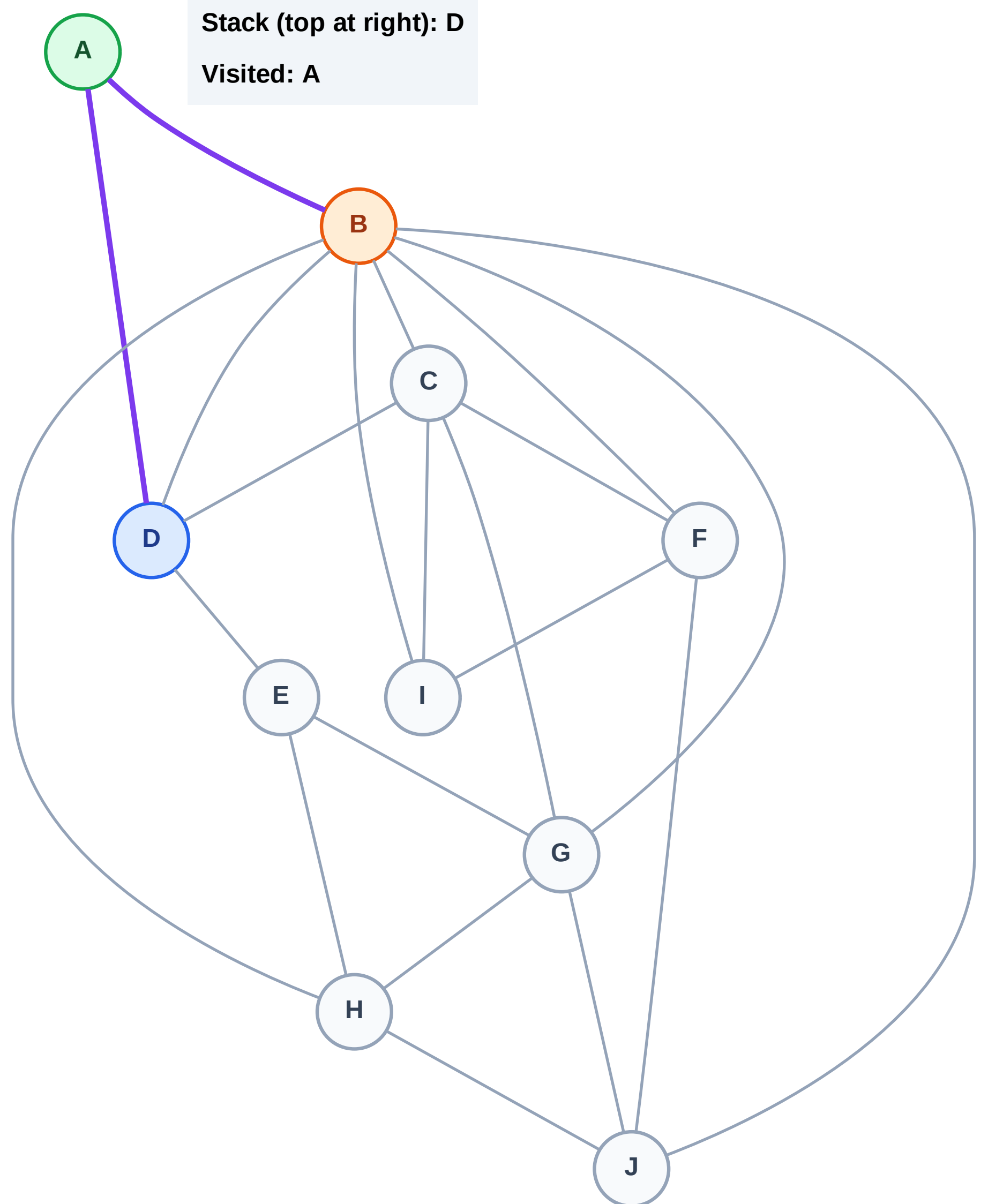
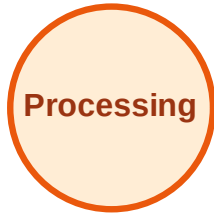
Legend



DFS Traversal

Step 4: Process node B

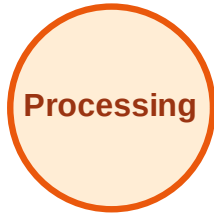
Legend



DFS Traversal

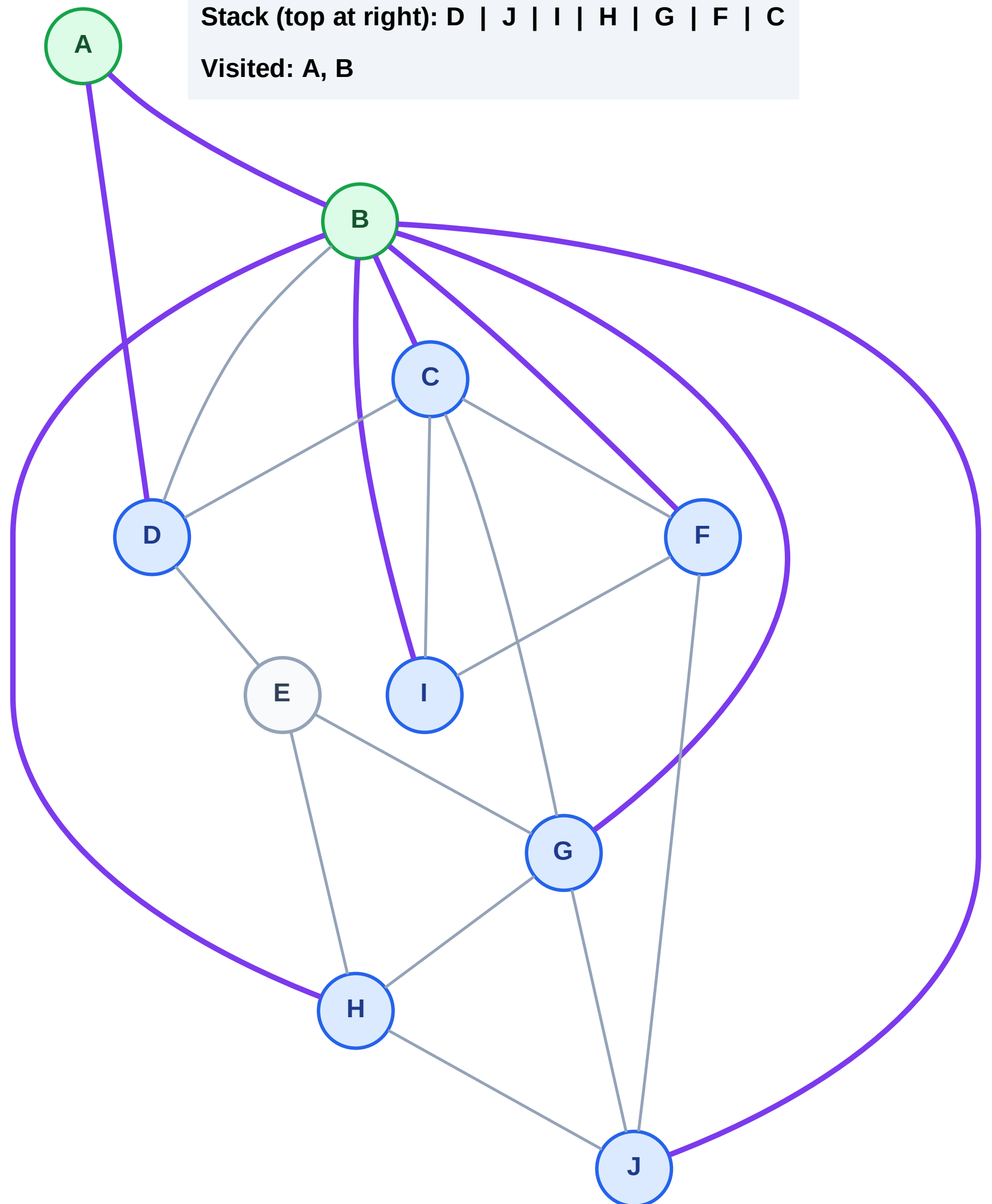
Step 5: Discover J, I, H, G, F, C from B

Legend



Stack (top at right): D | J | I | H | G | F | C

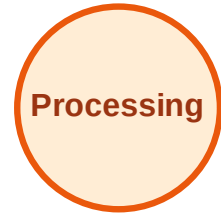
Visited: A, B



DFS Traversal

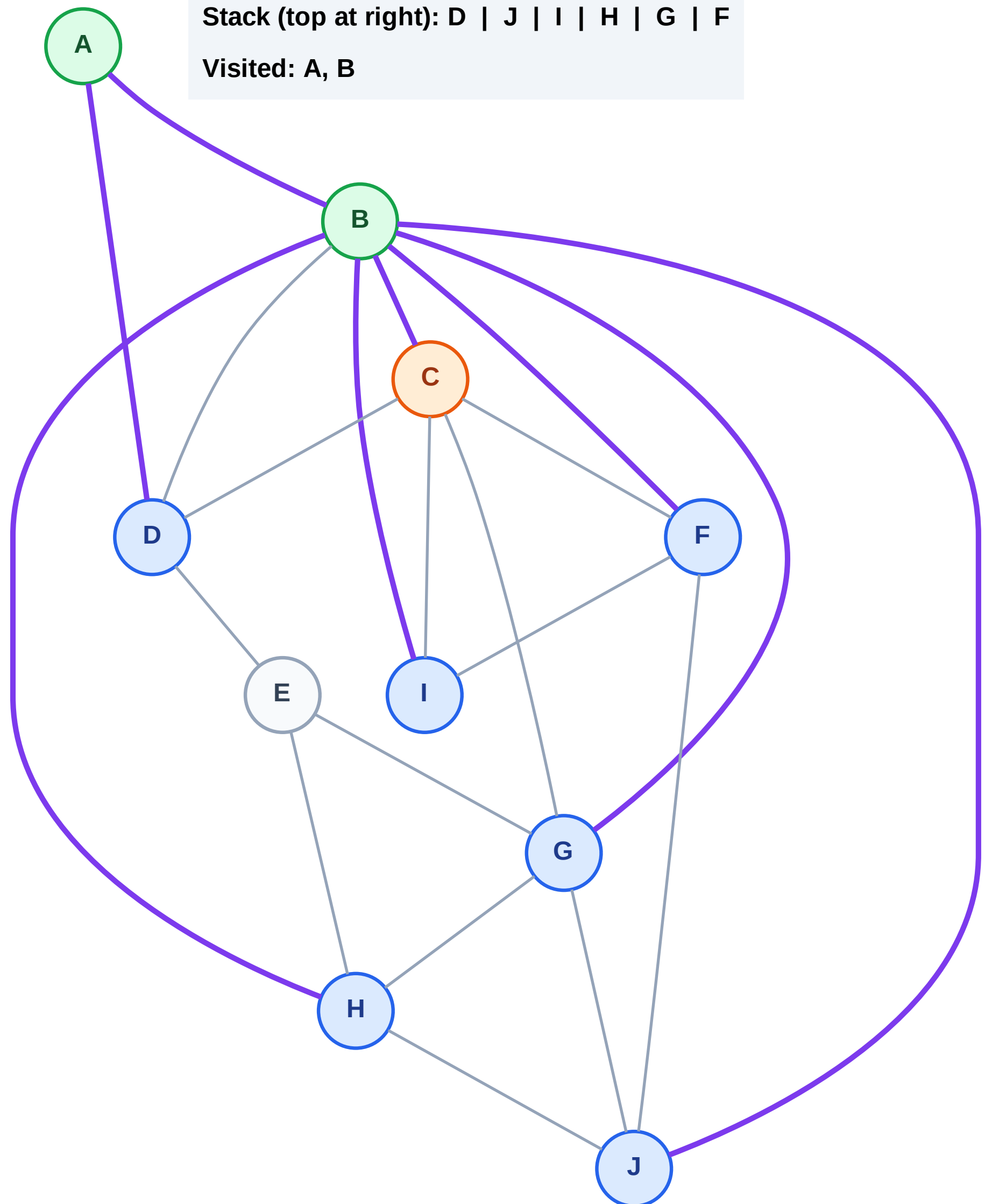
Step 6: Process node C

Legend



Stack (top at right): D | J | I | H | G | F

Visited: A, B



DFS Traversal

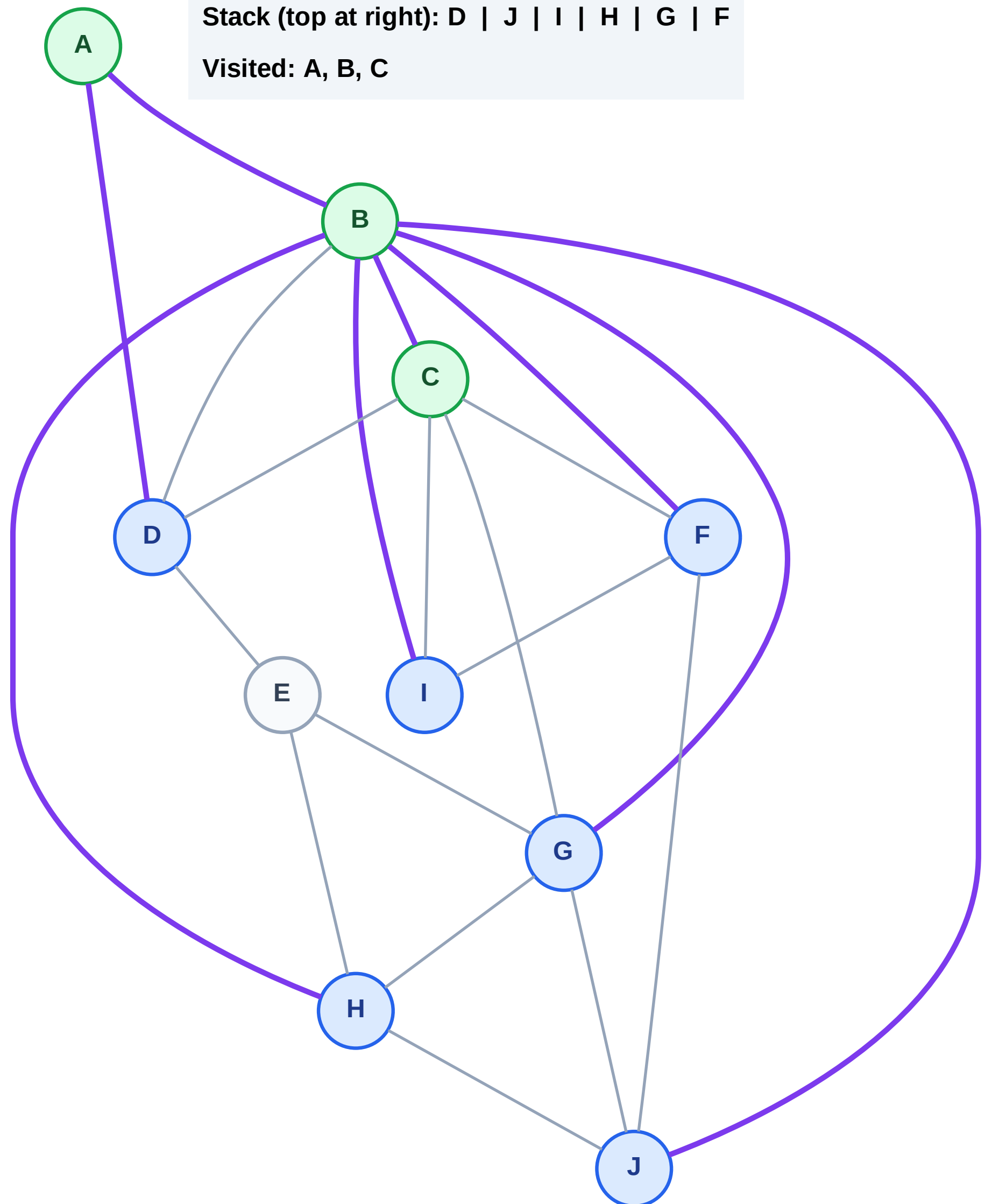
Step 7: No new neighbors from C

Legend



Stack (top at right): D | J | I | H | G | F

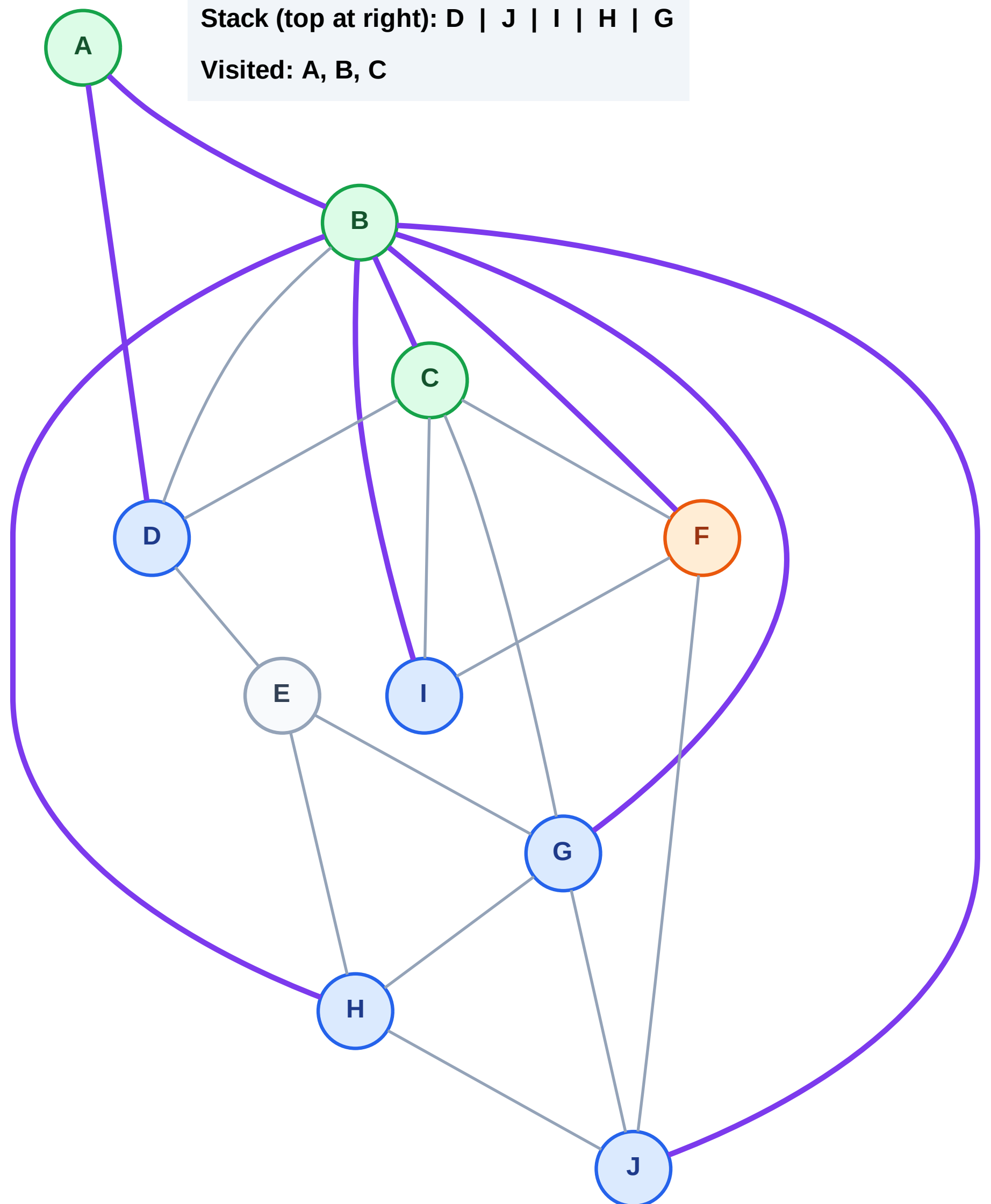
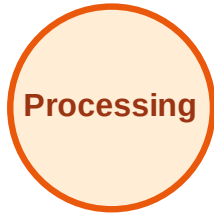
Visited: A, B, C



DFS Traversal

Step 8: Process node F

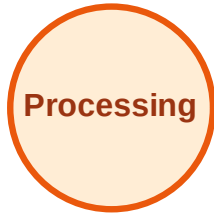
Legend



DFS Traversal

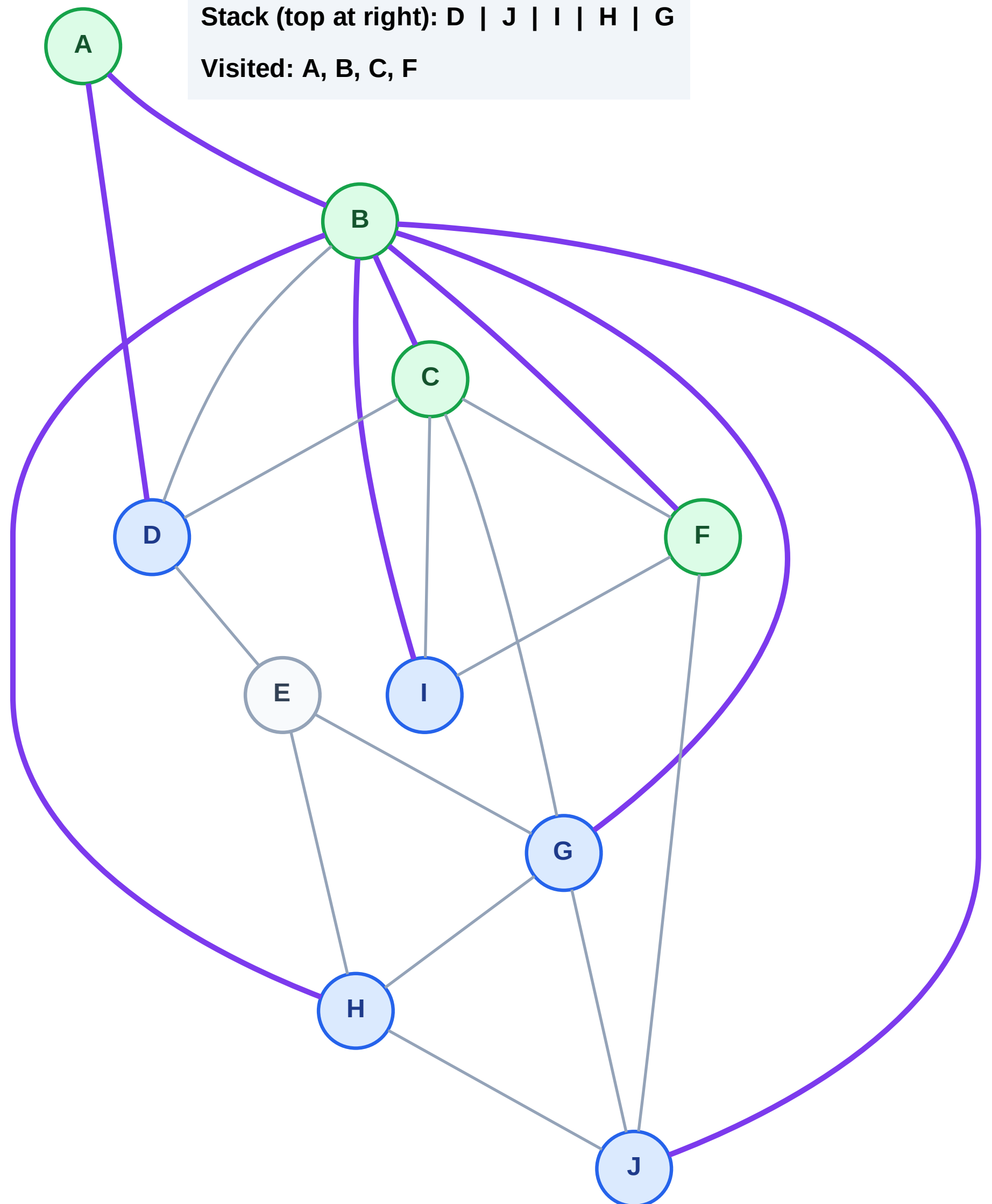
Step 9: No new neighbors from F

Legend



Stack (top at right): D | J | I | H | G

Visited: A, B, C, F



DFS Traversal

Step 10: Process node G

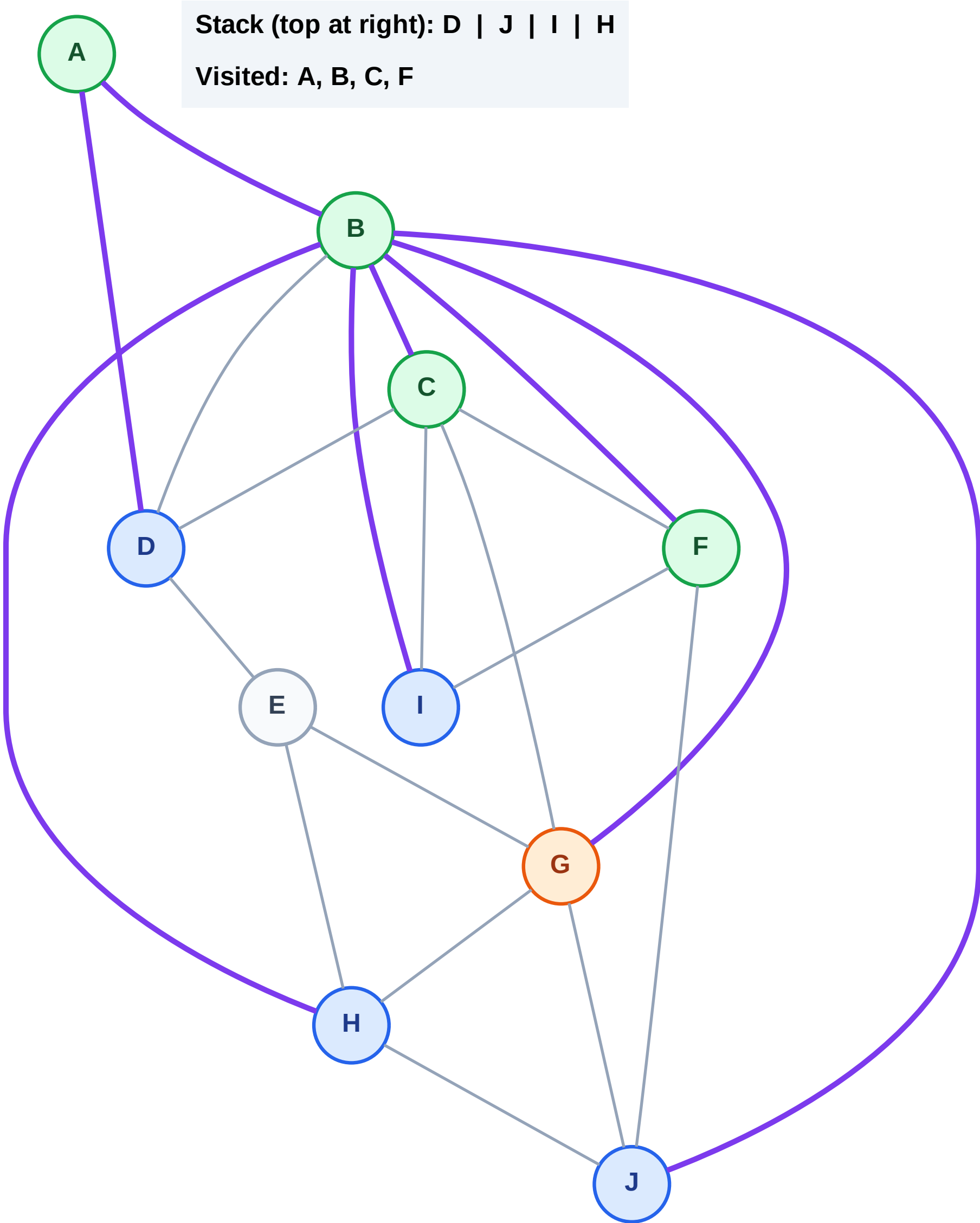
Legend

Complete

Processing

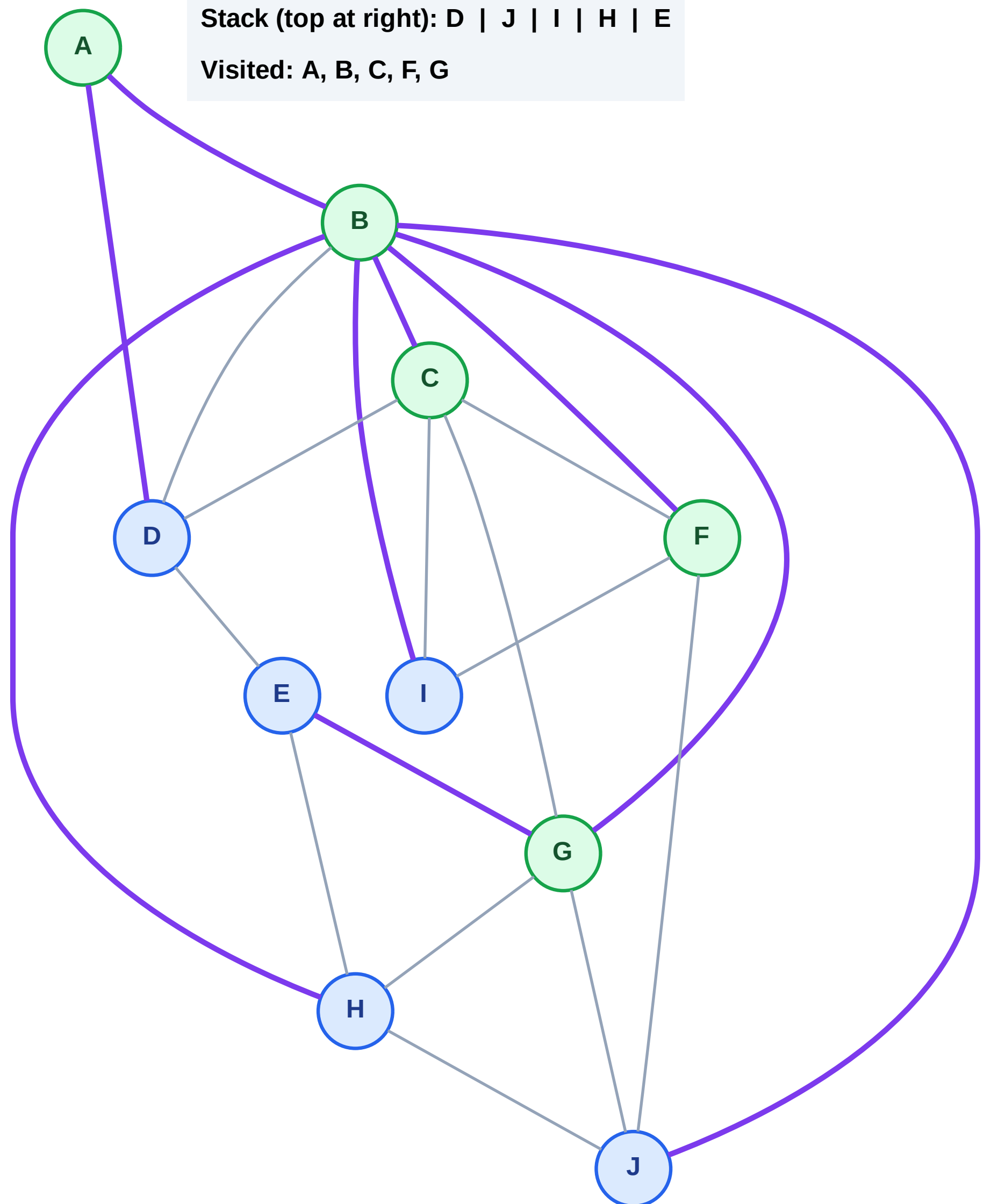
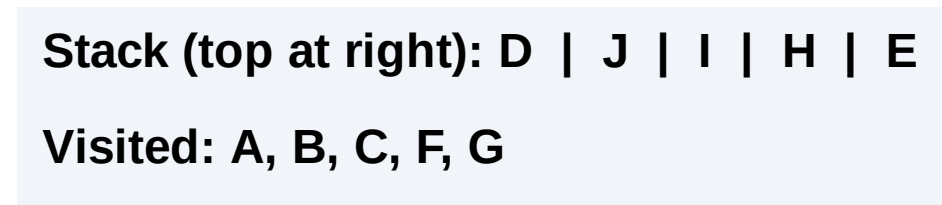
Discovered

Unseen



Step 11: Discover E from G

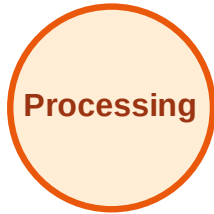
Step 11: Discover E from G



DFS Traversal

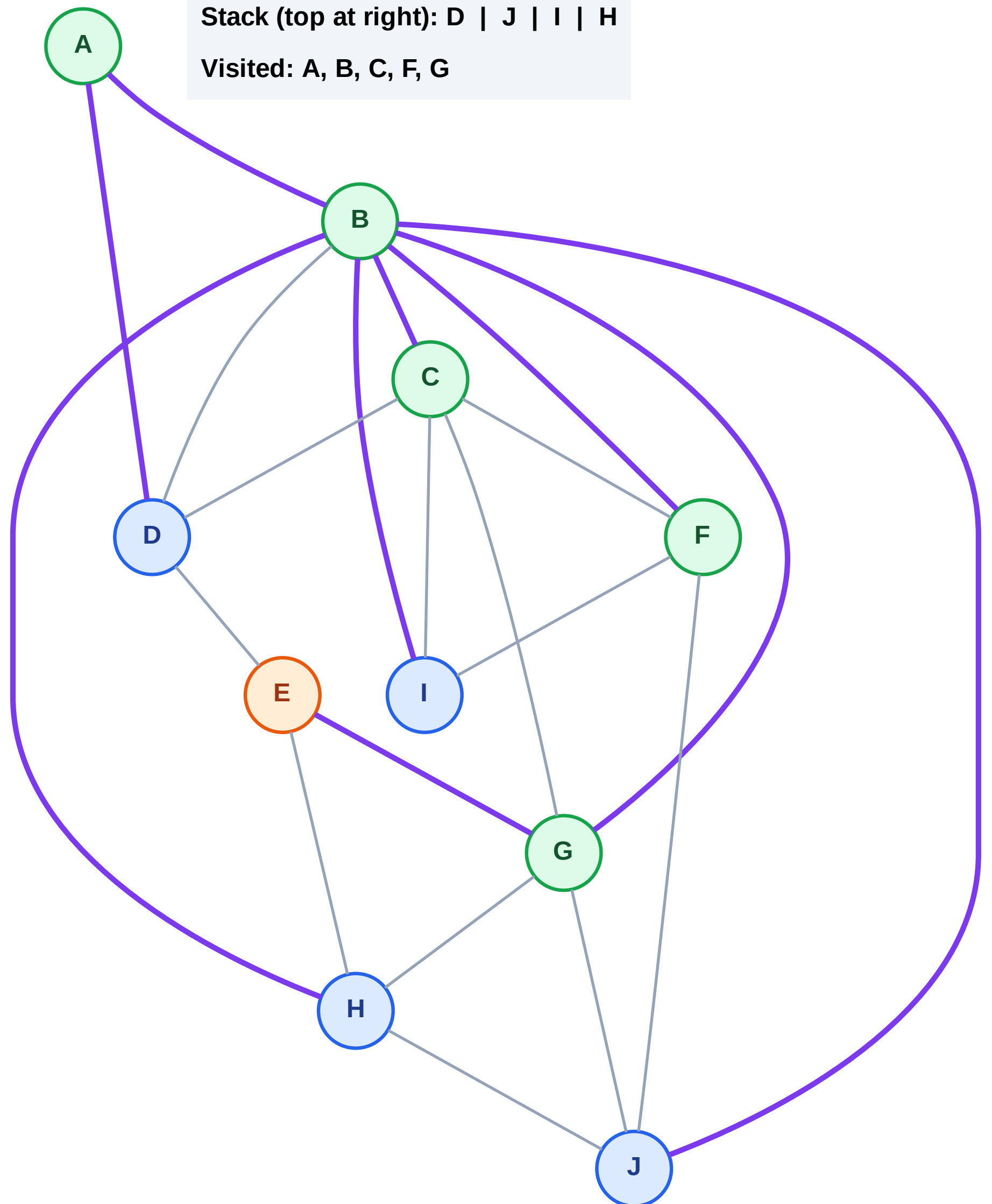
Step 12: Process node E

Legend

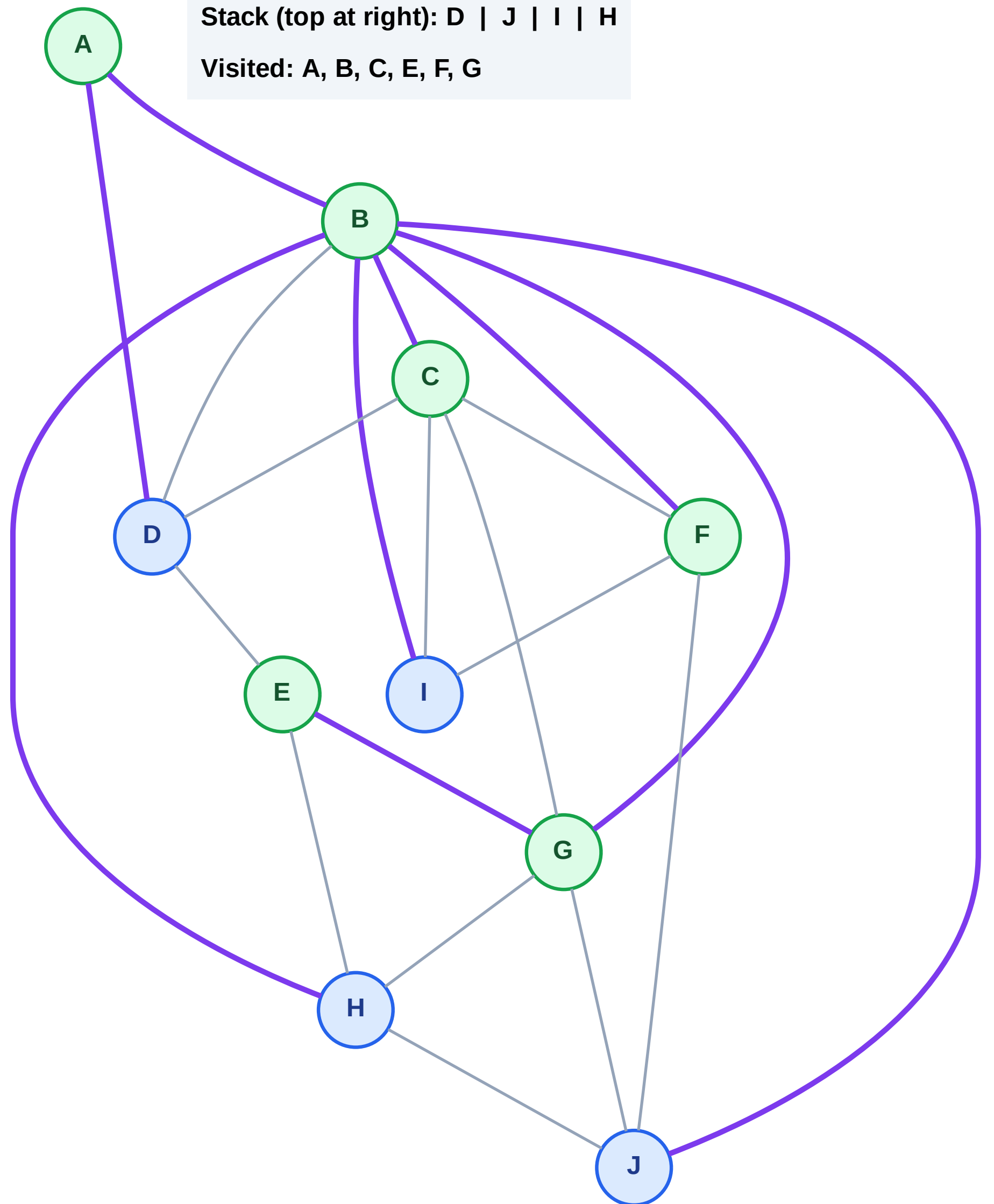
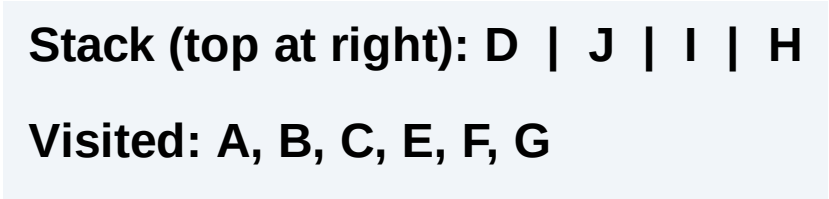


Stack (top at right): D | J | I | H

Visited: A, B, C, F, G



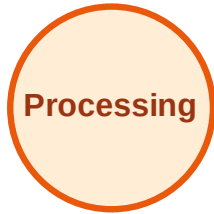
Step 13: No new neighbors from E



DFS Traversal

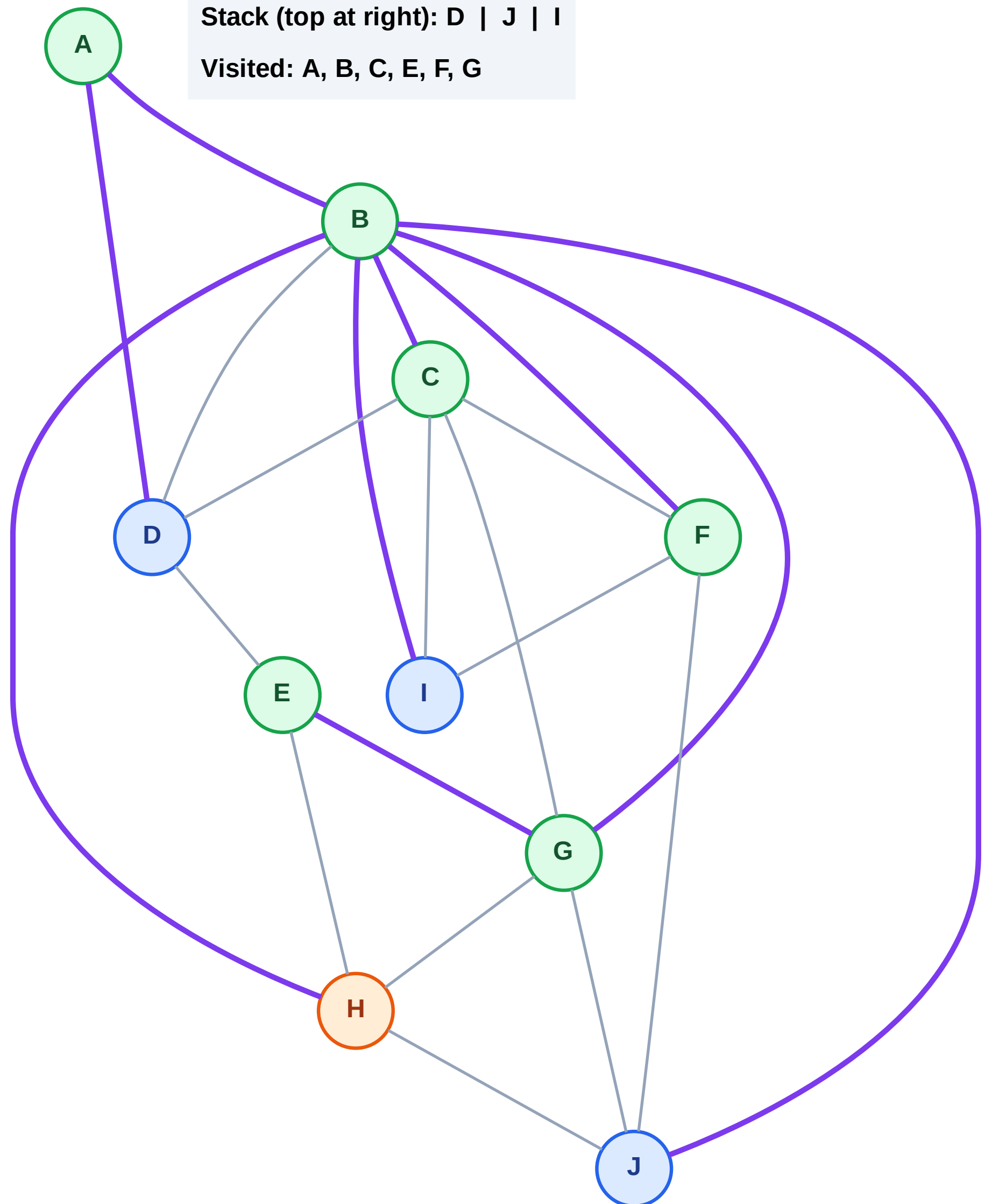
Step 14: Process node H

Legend



Stack (top at right): D | J | I

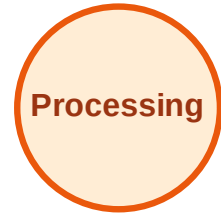
Visited: A, B, C, E, F, G



DFS Traversal

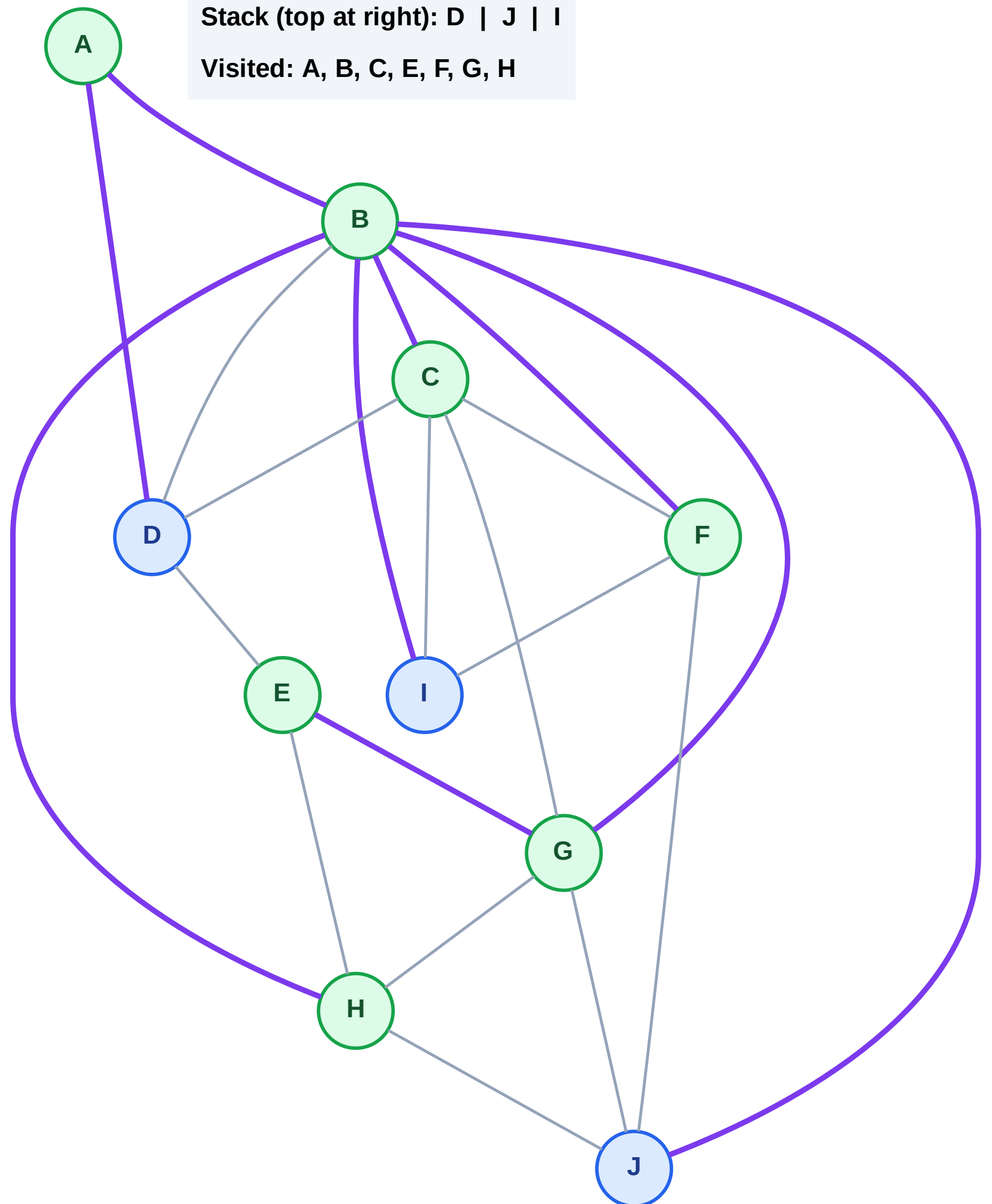
Step 15: No new neighbors from H

Legend



Stack (top at right): D | J | I

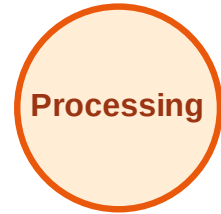
Visited: A, B, C, E, F, G, H



DFS Traversal

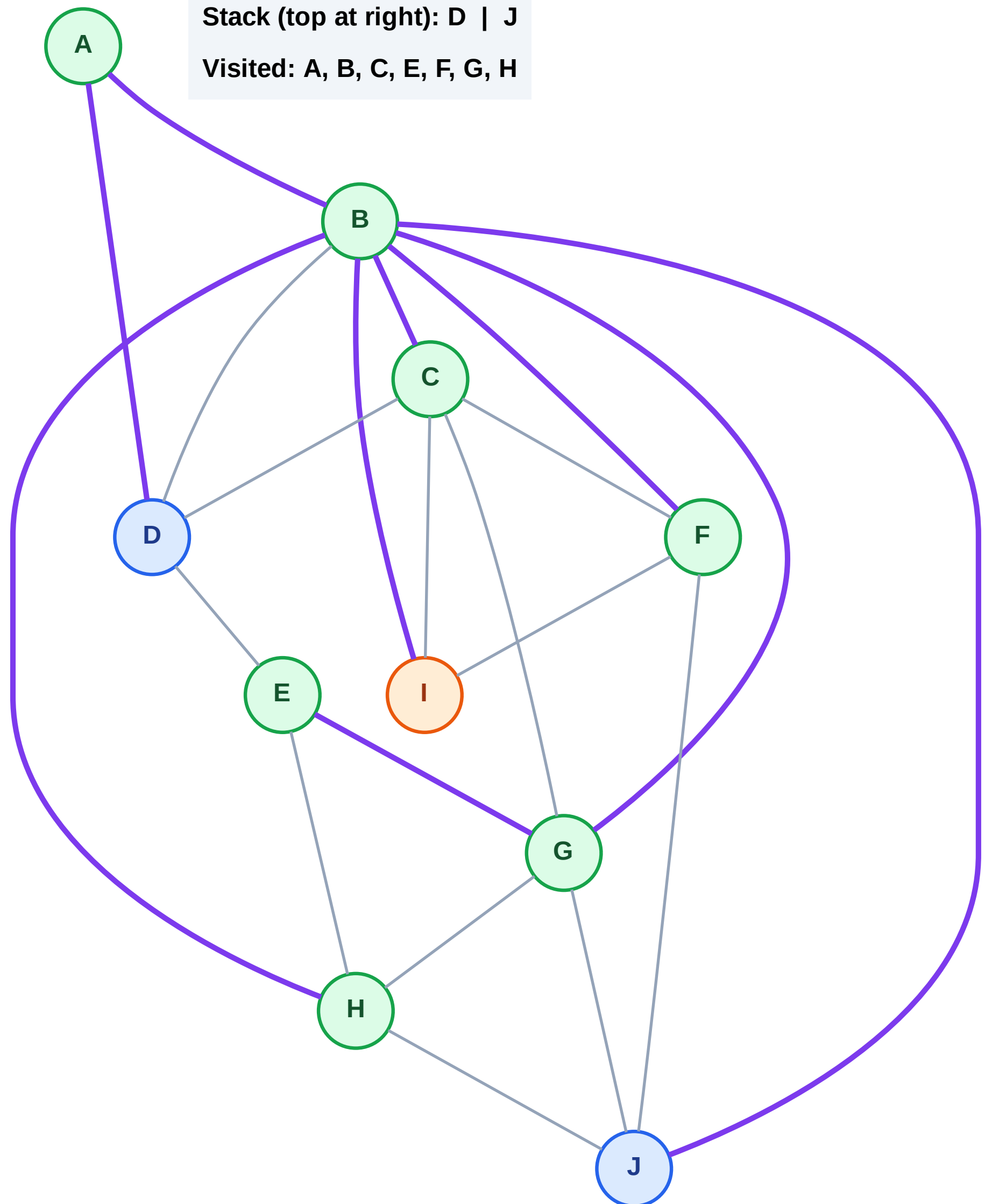
Step 16: Process node I

Legend



Stack (top at right): D | J

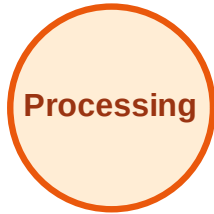
Visited: A, B, C, E, F, G, H



DFS Traversal

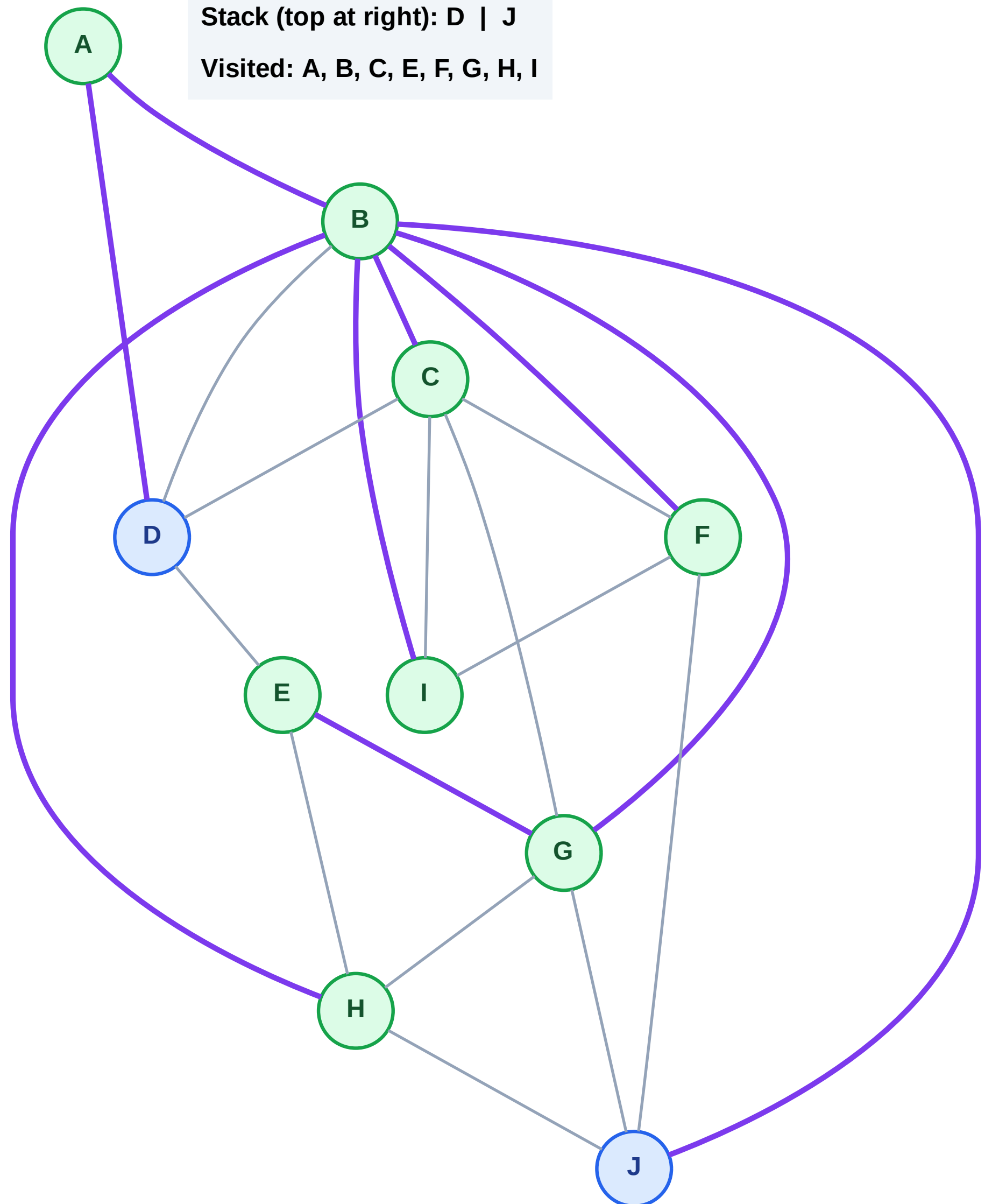
Step 17: No new neighbors from I

Legend



Stack (top at right): D | J

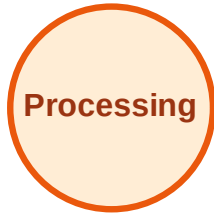
Visited: A, B, C, E, F, G, H, I



DFS Traversal

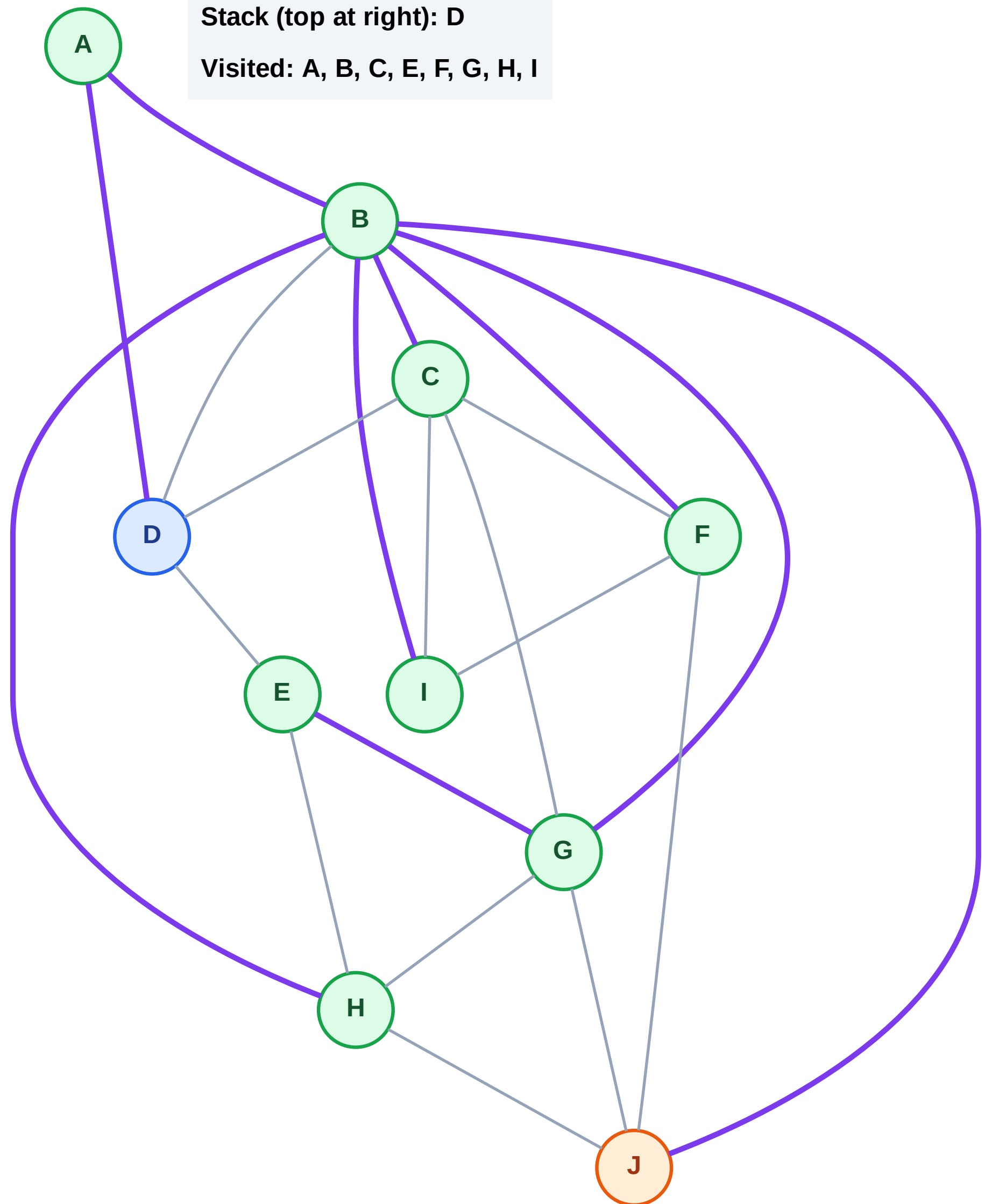
Step 18: Process node J

Legend



Stack (top at right): D

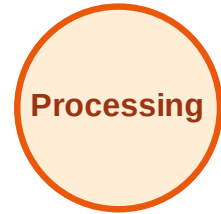
Visited: A, B, C, E, F, G, H, I



DFS Traversal

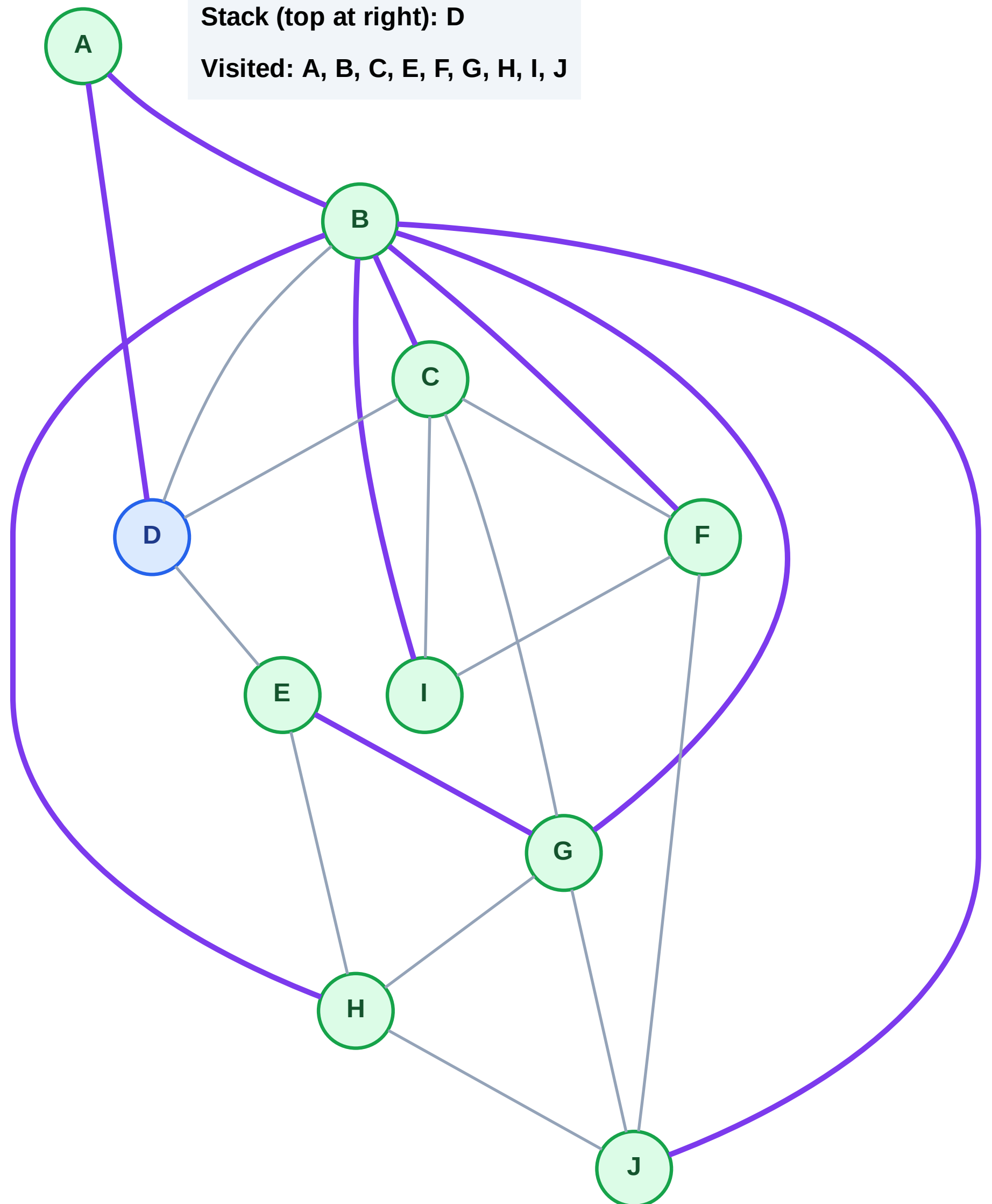
Step 19: No new neighbors from J

Legend



Stack (top at right): D

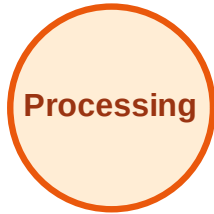
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

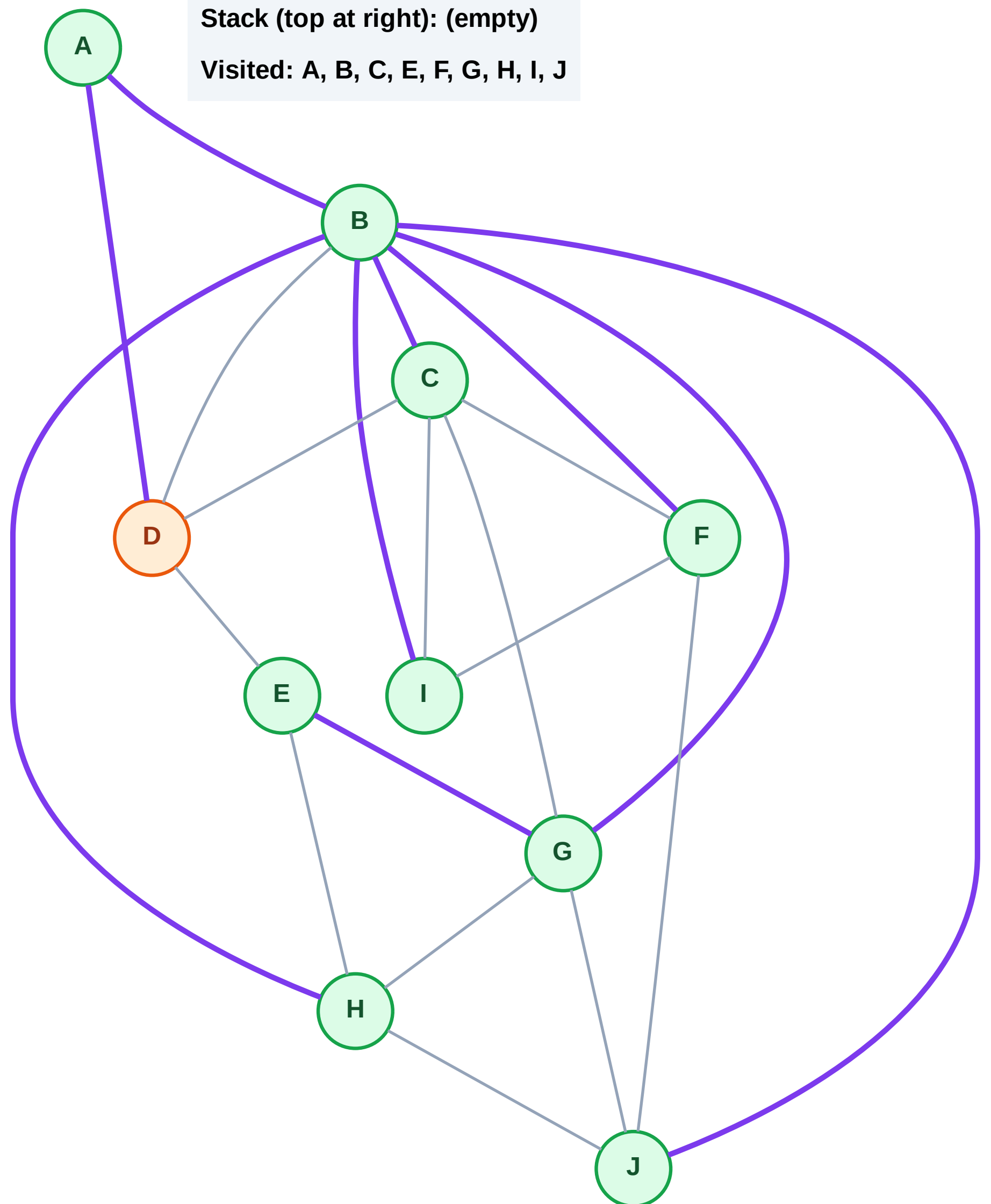
Step 20: Process node D

Legend



Stack (top at right): (empty)

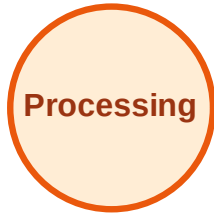
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from D

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

